

SpectralZero: 文本驱动的光谱-空间 对齐用于零样本高光谱图像分类

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摘要——近年来，基于深度学习的高光谱图像分类方法取得了显著进展。然而，现有方法主要集中于在训练阶段对可见类进行特征建模，而对不可见土地覆盖类别的泛化和迁移机制的探索仍然有限。因此，这些方法在零样本场景中表现出较差的适应性和泛化性能。为应对这一挑战，我们提出了一种名为 SpectralZero 的新型零样本高光谱图像分类框架，该框架利用大语言模型的强大泛化能力来弥补不可见类别中语义信息的缺失。此外，我们的框架显式捕获空间与光谱特征之间的协同关系，以促进从可见类到不可见类的知识迁移。具体而言，所提出的框架包含两个关键模块：语义提示增强和光谱-空间联合提取。语义提示增强利用大语言模型生成土地覆盖类别的细粒度语义描述，缓解因不可见类别缺少样本而导致的语义表示缺失。光谱-空间联合提取引入两个独立分支来分别建模空间结构和光谱反射特性，从而增强模型提取联合空间-光谱表示的能力。我们方法的一个主要优势在于，无需在训练或推理过程中使用任何样本即可准确分类不可见类别。在多个公开高光谱图像数据集上的实验表明，我们的方法在零样本

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在分类任务中展现出卓越的识别精度和泛化性能。代码将发布于<https://github.com/xiayang124/SpectralZero>

索引词——跨模态对齐，高光谱图像分类，零样本分类。

术语表

D 高光谱图像数据集 D，包含图像 I 和标签集 L。I 高光谱图像 $I \in \mathbb{R}^{H \times W \times C}$ 。H 高光谱图像的高度。W 高光谱图像的宽度。C 光谱波段的数量。 L^N 所有类别标签的集合， $L^N = \{l^1, l^2, \dots, l^N\}$ 。p 图块尺寸；每个图块的大小为 $p \times p$ 。 I^p 从 I 裁剪的图像块， $I^p \in \mathbb{R}^{p \times p \times C}$ 。S 可见类。U 不可见类。 I_S^p 来自可见类子集的高光谱图像块。 I_U^p 来自不可见类子集的高光谱图像块。 D^p 根据 D 构建的图块级数据集， $D^p = \{I^p, L\}$ 。 D_S^p 可见类子集， $D_S^p = \{I_S^p, L_S^K\}$ 。 D_U^p 不可见类子集， $D_U^p = \{I_U^p, L_U^M\}$ 。K 可见类的数量。M 不可见类的数量。 L_S^K 可见类的标签集， $L_S^K = \{l_S^1, l_S^2, \dots, l_S^K\}$ 。 L_U^M 不可见类的标签集， $L_U^M = \{l_U^1, l_U^2, \dots, l_U^M\}$ 。f_i 预训练的 CLIP 文本编码器。 o^t 由 f_i 提取的文本语义特征。 o^{spe} 由光谱分支提取的光谱特征。 o^{spa} 由 SSUE 中空间分支提取的空间特征。 S^{spe} 光谱跨模态嵌入空间。 S^{spa} 空间跨模态嵌入空间。 M^{spe} 光谱嵌入空间中的相似度矩阵。 M^{spa} 空间嵌入空间中的相似度矩阵。

SpectralZero: Text-Driven Spectral–Spatial Alignment for Zero-Shot Hyperspectral Image Classification

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Abstract—Recently, deep learning-based hyperspectral image (HSI) classification methods have witnessed significant advancements. However, existing approaches predominantly focus on feature modeling for seen classes during the training phase, while the exploration of generalization and transfer mechanisms toward unseen land-cover categories remains limited. As a result, these methods exhibit poor adaptability and generalization performance in zero-shot scenarios. To address this challenge, we propose a novel zero-shot HSI classification framework called SpectralZero that leverages the strong generalization capability of large language models (LLMs) to compensate for the absence of semantic information in unseen classes. Moreover, our framework explicitly captures the synergistic relationship between spatial and spectral features to facilitate knowledge transfer from seen to unseen categories. Specifically, the proposed framework comprises two key modules: semantic prompt enrichment (SPE) and spectral–spatial union extraction (SSUE). SPE utilizes LLMs to generate fine-grained semantic descriptions of land-cover classes, mitigating the semantic representation deficiency caused by the lack of samples in unseen categories. SSUE introduces two separate branches to independently model spatial structures and spectral reflectance characteristics, thereby enhancing the model’s capacity to extract joint spatial–spectral representations. A major advantage of our method lies in its ability to accurately classify unseen classes without requiring any samples during training or inference. Experiments on multiple public HSI datasets demonstrate that our approach consistently outperforms state-of-the-art methods in zero-shot

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classification tasks, exhibiting superior recognition accuracy and generalization performance. The code will be available at <https://github.com/xiayang124/SpectralZero>

Index Terms—Cross-modal alignment, hyperspectral image (HSI) classification, zero-shot classification.

NOMENCLATURE	
D	HSI dataset D with image I and label set L .
I	HSI $I \in \mathbb{R}^{H \times W \times C}$.
H	Height of the HSI.
W	Width of the HSI.
C	Number of spectral bands.
L^N	Set of all class labels, $L^N = \{l^1, l^2, \dots, l^N\}$.
p	Patch size; each patch is of size $p \times p$.
I^p	Image patch cropped from I , $I^p \in \mathbb{R}^{p \times p \times C}$.
S	Seen classes.
U	Unseen classes.
I_S^p	HSI patch from the seen classes subset.
I_U^p	HSI patch from the unseen classes subset.
D^p	Patch-level dataset constructed from D , $D^p = \{I^p, L\}$.
D_S^p	Seen classes subset, $D_S^p = \{I_S^p, L_S^K\}$.
D_U^p	Unseen classes subset, $D_U^p = \{I_U^p, L_U^M\}$.
K	Number of seen classes.
M	Number of unseen classes.
L_S^K	Label set of seen classes, $L_S^K = \{l_S^1, l_S^2, \dots, l_S^K\}$.
L_U^M	Label set of unseen classes, $L_U^M = \{l_U^1, l_U^2, \dots, l_U^M\}$.
f_i	Pretrained CLIP text encoder.
o^t	Textual semantic feature extracted by f_i .
o^{spe}	Spectral feature extracted by the spectral branch.
o^{spa}	Spatial feature extracted by the spatial branch in SSUE.
S^{spe}	Spectral cross-modal embedding space.
S^{spa}	Spatial cross-modal embedding space.
M^{spe}	Similarity matrix in the spectral embedding space.
M^{spa}	Similarity matrix in the spatial embedding space.

M 对 M^{spe} 和 M^{spa} 进行加权融合后的统一相似度矩阵。 M_{ij} (3) 式的分母中第 j 个相似度得分。 p_i^{cls} 光谱-空间联合特征。 p^{cls} 用于已知类别预测的置信度（概率）向量。 o^{spe} p 光谱-文本空间中的投影光谱特征。 o^{spa} p 空间-文本空间中的投影空间特征。 o^t p 对齐空间中的投影文本特征。 W^{spe} 光谱投影矩阵。 W^{spa} 空间投影矩阵。 W_t 文本投影矩阵。

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高 引言

光谱图像 (HSI) 提供了丰富的光

谱和空间信息，与传统的多光谱或RGB图像相比，能够更优越地识别土地覆盖类型 [1]。其连续的高分辨率光谱波段能够有效区分视觉上相似的类别，例如合成草地和天然草地，从而提升分类准确性和可靠性。HSI分类促进了联合的光谱-空间特征表示，支持水体分类 [2]、植物监测 [3]，和矿物识别 [4]等应用。然而，HSI的高维性给特征提取和分类带来了重大挑战，使得有效利用此类数据成为研究的关键重点。

早期高光谱图像分类方法主要依赖于人工设计的光谱-空间特征提取技术，采用经典机器学习算法如支持向量机 [5]、随机森林[6]，和K近邻 [7]。然而，这些受限于专家启发式的方法往往无法产生最优表征。随着深度学习的兴起，神经网络展现出强大的自动化特征学习能力，显著提升了分类性能。Chen等人 [8] 引入三维卷积神经网络来联合利用光谱和空间信息，取得了优异成果。Zhong等人[9]通过更深层次的架构和残差连接增强了模型稳定性和表现力。近期，Hong等人 [10] 采用Transformer的全局注意力机制对长程光谱依赖进行建模，有效解决了传统卷积模型的局部性局限，进一步提升了分类准确率。

尽管高光谱图像分类取得了显著进展，现有模型仍然严重依赖大量标注数据进行训练和评估。在实际应用中，单幅高光谱图像场景往往覆盖广阔的地理区域，并包含多样且复杂的土地覆盖类型。人工标注的过高成本使得全面标注不可行，导致某些类别的标注样本极度稀缺

甚至完全缺失 [11]。此外，地质活动或生物演化引起的地表变化，尽管能被传感器有效捕捉，但由于长期人工标注所需的高昂劳动力，难以进行持续性标注，从而加剧了零样本场景的出现。在这种情况下，传统分类模型往往难以准确建模不可见类的特征表示，导致其被误分类为已知类别，显著降低了整体分类性能和泛化能力。因此，开发在样本有限甚至零样本条件下仍具鲁棒性的高光谱图像分类方法，已成为一项迫切且关键的研究挑战。

为缓解标注成本高昂的问题，研究人员提出了多种小样本高光谱图像分类方法。小样本分类的目标是仅依赖极少量标注的高光谱图像样本（例如每类仅十个训练样本）即可实现可观的分类性能。Liu等人 [12] 构建了一个三维卷积神经网络来提取高光谱图像特征并学习度量空间，在测试阶段将其与最近邻分类器相结合执行高光谱图像分类，在小样本场景下取得了显著的性能提升。为降低有限数据条件下模型过拟合的风险，Xue等人[13]提出了更轻量化的网络架构，并引入不同尺度的正负样本来增强模型的泛化能力。此外，考虑到自然图像（如ImageNet[14]）比高光谱图像拥有丰富得多的标注数据，Li等人 [15] 首先利用自然图像预训练通用空间特征提取网络，随后使用有限的高光谱图像样本训练专用光谱分类分支。然而，这些方法仍不足以有效应对某些类别在训练标签中完全缺失的情况。

近年来，最初在 [16] 中提出的零样本分类——通过基于语言的属性识别不可见类别——日益受到关注。在该范式中，训练和测试的标签空间互斥，其中可见类代表训练期间可用的类别，而不可见类则表示模型从未见过但可通过学习到的语义关系推断出的新类别。主流方法利用 ImageNet [14]，CUB [17]，和AwA [18]，等图文数据集，将视觉和文本特征投影到共享的语义空间，以实现跨模态对齐。代表性方法包括语义输出编码 [19]，（将图像映射到语义嵌入以进行最近邻分类）以及基于深度学习的方法 [20]，[21]（利用大规模数据学习复杂的语义关系）。诸如基于生成对抗网络的框架 [22] 等生成模型通过为不可见类合成特征，进一步提升了泛化能力。然而，这些模型通常依赖有限的训练数据和小规模架构，限制了其语义建模能力。随着GPU计算的进步和大规模预训练模型的兴起，传统数据集日益显得

M	Unified similarity matrix after weighted fusion of M^{spe} and M^{spa} .
M_j	j th similarity score in the denominator of (3).
o^i	Spectral–spatial joint feature.
p^{cls}	Confidence (probability) vector for seen class prediction.
o^{spe}_p	Projected spectral feature in the spectral–text space.
o^{spa}_p	Projected spatial feature in the spatial–text space.
o^t_p	Projected text feature in the alignment space.
W_{spe}	Spectral projection matrix.
W_{spa}	Spatial projection matrix.
W_t	Text projection matrix.

I. INTRODUCTION

HYPER SPECTRAL images (HSI) offer rich spectral and spatial information, providing superior discrimination of land-cover types compared to traditional multispectral or RGB imagery [1]. Their continuous high-resolution spectral bands enable effective differentiation of visually similar classes, such as synthetic and natural grass, improving classification accuracy and reliability. HSI classification facilitates joint spectral–spatial feature representation, supporting applications like water body classification [2], plant monitoring [3], and mineral identification [4]. However, the high dimensionality of HSI poses significant challenges for feature extraction and classification, making efficient utilization of such data a key research focus.

Early HSI classification methods primarily relied on manually designed spectral–spatial feature extraction techniques, employing classical machine learning algorithms such as support vector machines (SVMs) [5], random forests (RFs) [6], and K-nearest neighbors (K-NNs) [7]. However, these approaches, constrained by expert heuristics, often fail to yield optimal representations. With the advent of deep learning, neural networks have shown a remarkable capacity for automated feature learning, significantly advancing classification performance. Chen et al. [8] introduced a 3-D convolutional neural network (3D-CNN) to jointly exploit spectral and spatial information, achieving superior results. Zhong et al. [9] enhanced model stability and expressiveness through deeper architectures and residual connections. More recently, Hong et al. [10] adopted the transformer’s global attention mechanism to model long-range spectral dependencies, effectively addressing the locality limitations of conventional convolutional models and further boosting classification accuracy.

Despite the significant progress achieved in HSI classification, existing models still heavily rely on large amounts of labeled data for training and evaluation. In practical applications, a single HSI scene often covers extensive geographic areas and contains diverse and complex land-cover types. The prohibitively high cost of manual annotation renders comprehensive labeling infeasible, resulting in severe scarcity

or even complete absence of labeled samples for certain classes [11]. Moreover, surface variations induced by geological activities or biological evolution, though effectively captured by sensors, are difficult to annotate continuously due to the high labor demands of long-term manual labeling, thereby exacerbating the emergence of zero-shot scenarios. In such cases, traditional classification models often struggle to accurately model the feature representations of unseen classes, leading to their misclassification as known categories and significantly impairing overall classification performance and generalization. Consequently, developing HSI classification methods that are robust under limited-sample or even zero-shot conditions has become an urgent and critical research challenge.

To alleviate the issue of high annotation costs, researchers have proposed several limited-sample HSI classification methods. The goal of limited-sample classification is to achieve considerable classification performance but relying solely on very limited labeled HSI samples (e.g., as few as ten samples per class for training). Liu et al. [12] constructed a 3D-CNN to extract HSI features and learn a metric space, and combined it with a nearest-neighbor classifier at the testing stage to perform HSI classification, achieving significant performance improvement in limited-sample scenarios. To mitigate the risk of model overfitting under limited data conditions, Xue et al. [13] proposed a more lightweight network architecture and introduced positive and negative samples across different scales to enhance the generalization capability of the model. Moreover, considering that natural images (such as ImageNet [14]) possess substantially richer labeled data compared to HSI, Li et al. [15] first pretrained a general spatial feature extraction network using natural images, subsequently training a dedicated spectral classification branch with limited HSI samples. Nevertheless, these methods remain inadequate for effectively addressing situations in which certain categories are entirely absent from the training labels.

In recent years, zero-shot classification—originally introduced in [16] to recognize unseen categories via language-based attributes—has garnered increasing attention. In this paradigm, the label spaces of training and testing are mutually exclusive, where the seen classes represent the categories available during training and the unseen classes denote novel categories that have never been observed by the model but can be inferred through learned semantic relationships. Mainstream approaches leverage image–text datasets such as ImageNet [14], CUB [17], and AwA [18], projecting visual and textual features into a shared semantic space to enable cross-modal alignment. Representative methods include semantic output codes [19], which map images into semantic embeddings for nearest-neighbor classification, and deep learning-based approaches [20], [21] that exploit large-scale data to learn complex semantic relations. Generative models such as GAN-based frameworks [22] further improve generalization by synthesizing features for unseen classes. However, these models often rely on limited training data and small-scale architectures, constraining their semantic modeling capacity. With advances in GPU computing and the rise of large-scale pretrained models, traditional datasets are increasingly

不足。为应对这一问题, CLIP [23]被提出, 它利用大规模网络来源的图文对和对比学习构建了一个统一语义空间, 此后该空间被扩展到语义分割 [24]、图像生成 [25], [26]、点云分类 [27]和异常检测 [28]等任务中。然而, 这些方法主要针对RGB图像开发, 缺乏捕捉高光谱图像中丰富光谱特征的能力, 限制了它们在该领域的泛化能力。

早期的几项研究已经探索了高光谱图像中的零样本分类。Yue等人 [31] 提出了一种光谱-空间重构方法, 以增强可见类与不可见类之间的区分能力。然而, 该方法将所有不可见类别视为单一群体, 缺乏细粒度的区分。Pang等人 [32] 引入了一种基于CLIP的伪标签框架, 该框架通过数据增强与精炼来减少噪声, 从而在无需人工标注的情况下提升稳定性。然而, 其对文本语义的利用仍十分有限, 制约了视觉-语义表征学习。为进一步挖掘文本信息固有的语义迁移能力, Zhang等人 [30] 提出使用文本描述作为辅助信息, 以促进不同数据集之间分类能力的迁移。尽管该方法有效利用了侧边信息, 但其在训练过程中仍依赖类别标签, 且仅限于跨数据集类别完全相同的场景, 因此难以扩展到零样本分类设置中。为克服对特定数据集的训练依赖, Li等人 [29]开发了HyperFree, 该方法利用全光谱词典和极简提示, 可在高光谱图像数据集间进行泛化。尽管HyperFree具有较高的有效性, 但其在推理时仍依赖类别特定提示, 因此在严格的零样本设置下无法识别不可见类别。图1展示了这些方法的对比。总之, 当前关于高光谱图像零样本分类的研究仍处于探索阶段, 并面临若干显著局限性。

- 1) 大多数主流零样本分类方法主要基于RGB图像进行训练, 未能充分建模和利用高光谱图像中丰富的光谱信息, 从而限制了它们在超光谱场景下的泛化能力。
- 2) 现有的高光谱图像零样本分类模型 (参见[32]) 对辅助信息——如文本语义或类别属性描述——的整合与利用不足, 因而未能充分发挥语义知识在增强类别区分和知识迁移方面的潜力。

为应对上述挑战, 本文提出了一种新颖的高光谱图像零样本分类框架, 称为语义驱动的光谱-空间对齐 (SpectralZero)。SpectralZero 通过利用来自可见类的标注样本及相关辅助语义信息, 有效识别未见土地覆盖类别, 从而解决因数据稀缺或标注成本高昂导致的局限性。该框架包含两个关键组件: 语义提示增强 (SPE) 模块和光谱-空间联合提取 (SSUE) 模块。SPE模块通过视觉属性描述丰富类别表示, 以增强语义泛化能力。SSUE模块引入了一种

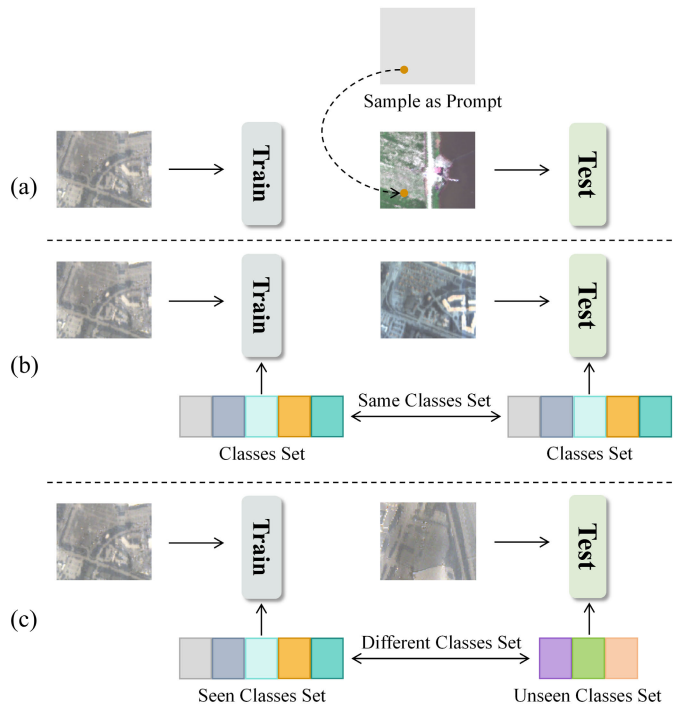


图1. 三种模型之间的区别。(a) 基于提示学习的零样本分类 (例如 HyperFree [29]), 仍需要与类别相关的样本作为测试提示; (b) 域泛化 (参见 [30]), 在同一数据集和类别设置中进行; (c) 文本引导的零样本分类 (本文方法), 其中训练和测试类别互不相交, 模型通过跨类别知识迁移识别不同的土地覆盖类型。

解耦光谱特征提取路径, 其中全局空间池化抑制了结构干扰, 双投影空间独立地从空间和光谱特征计算图像-文本相似性矩阵。这些矩阵通过比例加权融合形成用于对比学习的统一相似度矩阵, 从而改善跨模态对齐和未见类别分类。SPE模块与SSUE模块协同作用, 共同增强模型在语义建模和光谱表示方面的能力, 促进对新颖类别的稳健泛化。

- 1) 本文提出了一种名为SpectralZero的新型高光谱图像分类框架, 专为零样本场景设计。该框架利用大语言模型的泛化能力, 有效解决了未见类别信息缺失问题。此外, 它通过挖掘空间-光谱特征, 促进了从可见类到不可见类的知识迁移。在多个公开数据集上的实验结果表明, 所提框架优于现有零样本分类方法, 验证了其有效性和泛化能力。
- 2) 引入SPE模块, 基于大语言模型生成土地覆盖类别的细粒度语义描述, 缓解不可见类别视觉信息匮乏的问题。此外, 该模块通过CLIP模型将辅助语义信息嵌入统一语义空间, 从而构建可见类与不可见类之间的隐式语义关联。

inadequate. To address this, CLIP [23] was proposed, leveraging large-scale web-sourced image–text pairs and contrastive learning to construct a unified semantic space, which has since been extended to tasks like semantic segmentation [24], image generation [25], [26], point cloud classification [27], and anomaly detection [28]. Nonetheless, these approaches are primarily developed for RGB imagery and lack the capacity to capture rich spectral features in HSI, limiting their generalization in such domains.

Several early efforts have explored zero-shot classification in HSI. Yue et al. [31] proposed a spectral–spatial reconstruction method to enhance discrimination between seen and unseen classes. However, this approach treats all unseen categories as a single group, lacking fine-grained differentiation. Pang et al. [32] introduced a CLIP-based pseudo-labeling framework that mitigates noise through augmentation and refinement, improving stability without requiring manual annotation. Yet its exploitation of textual semantics remains limited, constraining visual-semantic representation learning. To further exploit the semantic transfer capability inherent in textual information, Zhang et al. [30] proposed using textual descriptions as auxiliary information to facilitate classification capability transfer between different datasets. Although this approach effectively leverages side information, it still relies on class labels during training and is restricted to scenarios where classes across datasets are identical, making it challenging to extend to zero-shot classification settings. To overcome dataset-specific training dependencies, Li et al. [29] developed HyperFree, which utilizes a spectral-wide dictionary and minimal prompts to generalize across HSI datasets. Despite its effectiveness, HyperFree still depends on class-specific prompts at inference, thus failing to identify unseen categories in strict zero-shot settings. A comparison of these approaches has been shown in Fig. 1. In summary, the current research on zero-shot classification of HSI remains at an exploratory stage and faces several notable limitations.

- 1) Most mainstream ZSC methods are primarily trained on RGB imagery, failing to fully model and exploit the rich spectral information inherent to HSIs, thereby limiting their generalization capability in hyperspectral scenarios.
- 2) Existing HSI-ZSC models (see [32]) exhibit inadequate integration and utilization of auxiliary information—such as textual semantics or class attribute descriptions—thus, underutilizing the potential of semantic knowledge in enhancing class discrimination and knowledge transfer.

To tackle the aforementioned challenges, this article proposes a novel framework for HSI zero-shot classification, termed semantic-driven spectral–spatial alignment (SpectralZero). SpectralZero enables effective recognition of unseen land-cover categories by leveraging labeled samples from seen classes and associated auxiliary semantics, addressing limitations due to data scarcity or annotation costs. The framework comprises two key components: the semantic prompt enrichment (SPE) module and the spectral–spatial union extraction (SSUE) module. SPE module enriches category representations with visual attribute descriptions to enhance semantic generalization. SSUE module introduces a

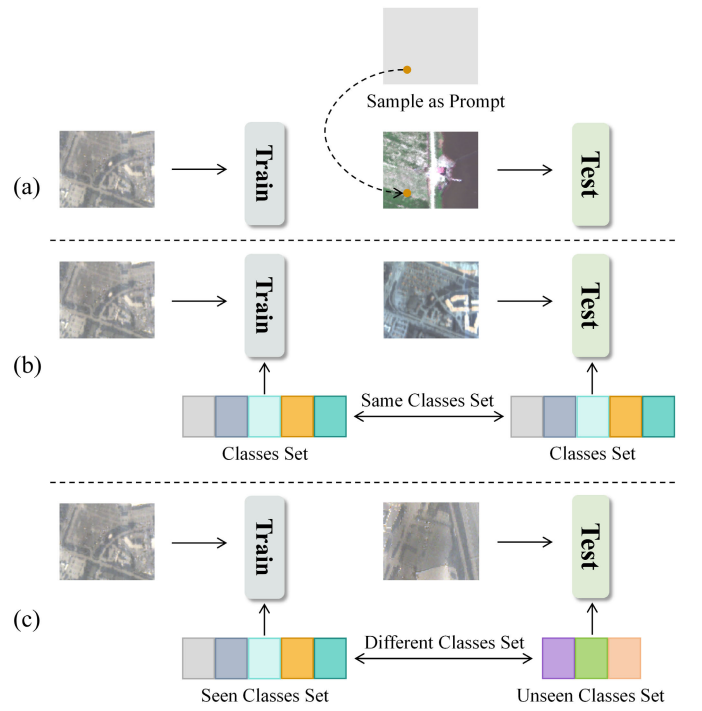


Fig. 1. Difference among three models. (a) Prompt learning-based zero-shot classification (such as HyperFree [29]), which still requires class-related samples as prompts for testing; (b) domain generalization (see [30]), which is conducted within the same dataset and category settings; and (c) text-guidance zero-shot classification (ours), where the train and test classes are disjoint, and the model identifies different land-cover categories through cross-class knowledge transfer.

decoupled spectral feature extraction pathway, where global spatial pooling suppresses structural interference, and dual projection spaces compute image–text similarity matrices from spatial and spectral features independently. These are fused via proportional weighting to form a unified similarity matrix for contrastive learning, thereby improving cross-modal alignment and unseen class classification. Together, the SPE module and SSUE module synergistically strengthen the model’s capacity for semantic modeling and spectral representation, facilitating robust generalization to novel categories.

- 1) This article proposes a novel HSI classification framework called SpectralZero tailored for zero-shot scenarios. By leveraging the generalization capability of large language models (LLMs), the framework effectively addresses the information deficiency of unseen classes. Furthermore, it exploits spatial–spectral features to facilitate knowledge transfer from seen to unseen categories. Experimental results on multiple publicly available datasets demonstrate that the proposed framework outperforms existing zero-shot classification methods, validating its effectiveness and generalizability.
- 2) An SPE module is introduced to generate fine-grained semantic descriptions of land-cover classes based on LLMs, mitigating the lack of visual information for unseen categories. In addition, the module embeds auxiliary semantic information into a unified semantic space via the CLIP model, thereby constructing implicit semantic associations between seen and unseen classes.

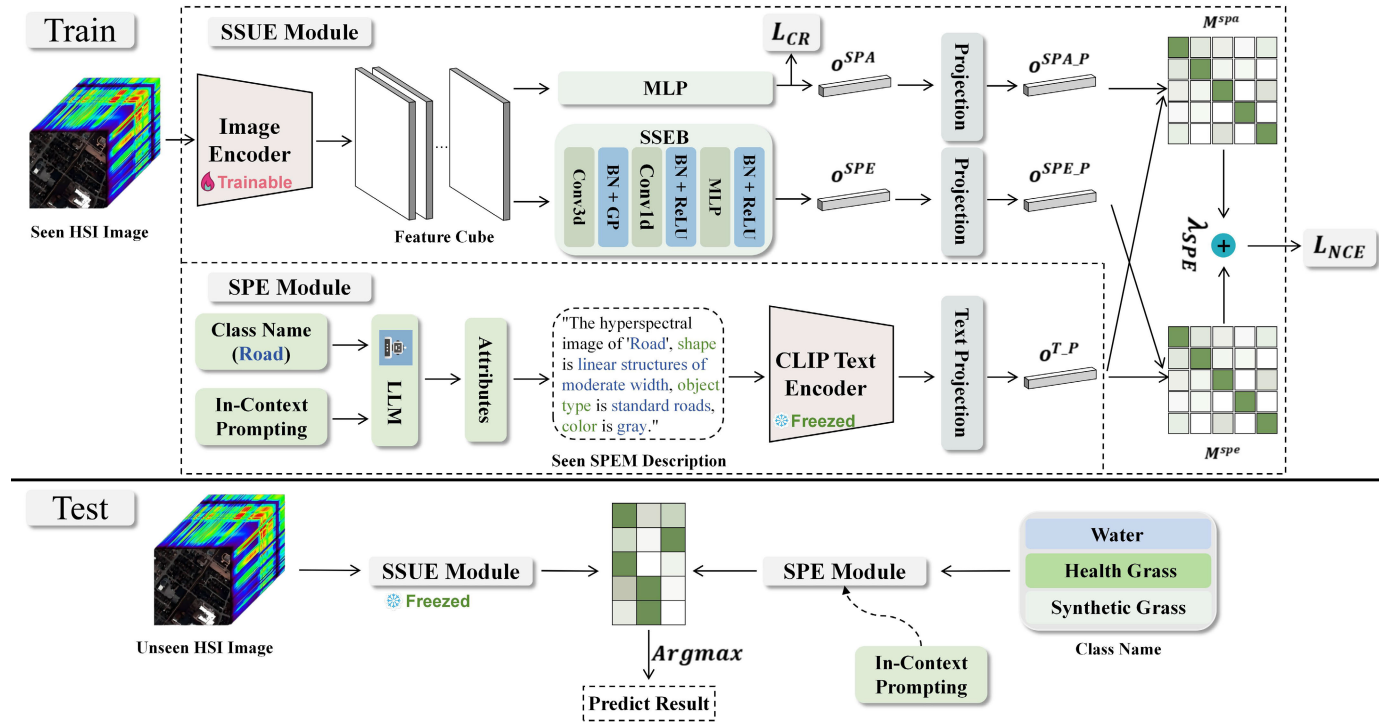


图2. 所提出的SpectralZero框架。SPE模块将丰富的自然语言描述编码为语义特征，并将其投影到空间和光谱嵌入空间中。SSUE模块将图像特征解耦为空间和光谱成分，然后与相应的语义特征进行对齐。两个对齐空间中的相似度矩阵被融合，并在训练期间用于计算损失。在测试时，通过选择融合相似度矩阵中最大值对应的类别索引来执行分类。

这显著增强了模型的语义建模与识别能力，为零样本分类任务提供了关键支持。

3) 设计了一个SSUE模块，以提高模型在零样本条件下提取空间和光谱特征的能力。该模块由两个独立的分支组成：一个空间分支，用于捕捉土地覆盖的结构信息；一个光谱分支，用于显式建模高光谱数据中固有的丰富反射特征。然后，这些特征被深度融合，生成一个能够联合表达空间和光谱属性（如颜色和材料属性）的统一嵌入，从而显著提升零样本场景中的分类性能。

II. 方法论

本节介绍所提出的SpectralZero框架的整体架构，并详细描述其两个核心组件：SPE模块和SSUE模块。此外，第II-D节进一步阐述了模型的实现细节。SpectralZero的整体流程如图2所示。

A. 概述

给定一个高光谱图像数据集 $D = \{\mathbf{I}, \mathbf{L}\}$ ，其中 $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ 表示一幅高光谱图像， H 和 W 分别代表图像的

高度和宽度，而 C 是光谱波段的数量。标签集表示为 $L^N = \{l^1, l^2, \dots, l^N\}$ 。为便于模型训练与评估，我们从数据集 D 的图像 $\mathbf{I}$ 中，以特定像素为中心提取大小为 $p \times p$ 的图像块，并将其与对应的类别标签关联，以构建子集 $D^p = \{\mathbf{I}^p, \mathbf{L}\}$ ，其中 $\mathbf{I}^p \in \mathbb{R}^{p \times p \times C}$ 。根据零样本分类设置，子集 D^p 进一步划分为可见类子集 $D_S^p = \{\mathbf{I}_S^p, \mathbf{L}_S^K\}$ 和不可见类子集 $D_U^p = \{\mathbf{I}_U^p, \mathbf{L}_U^M\}$ ，其中 S 和 U 分别表示可见类和不可见类， K 和 M 分别表示可见类和不可见类的数量。来自可见类子集的每个样本包含一个图像块 $\mathbf{I}_S^p \in \mathbb{R}^{p \times p \times C}$ 、一个标签集 $\mathbf{L}_S^K = \{l_S^1, l_S^2, \dots, l_S^K\} \in L^N$ 。不可见类子集的构建与可见类子集类似，包括图像块 $\mathbf{I}_U^p \in \mathbb{R}^{p \times p \times C}$ 和不可见标签集 $\mathbf{L}_U^M = \{l_U^1, l_U^2, \dots, l_U^M\} \in L^N$ 。

本文聚焦于严格零样本分类设定，其中可见类和不可见类的标签集不相交，且满足

$$\mathbf{L}_S^K \cap \mathbf{L}_U^M = \emptyset. \quad (1)$$

严格零样本分类的目标是确保没有未见类别的样本参与训练。在此约束下，仅使用可见类子集 D_S^p 作为训练数据来学习一个映射函数

$$f: \mathbf{I}_S^p \rightarrow \mathbf{I}_U^p \quad (2)$$

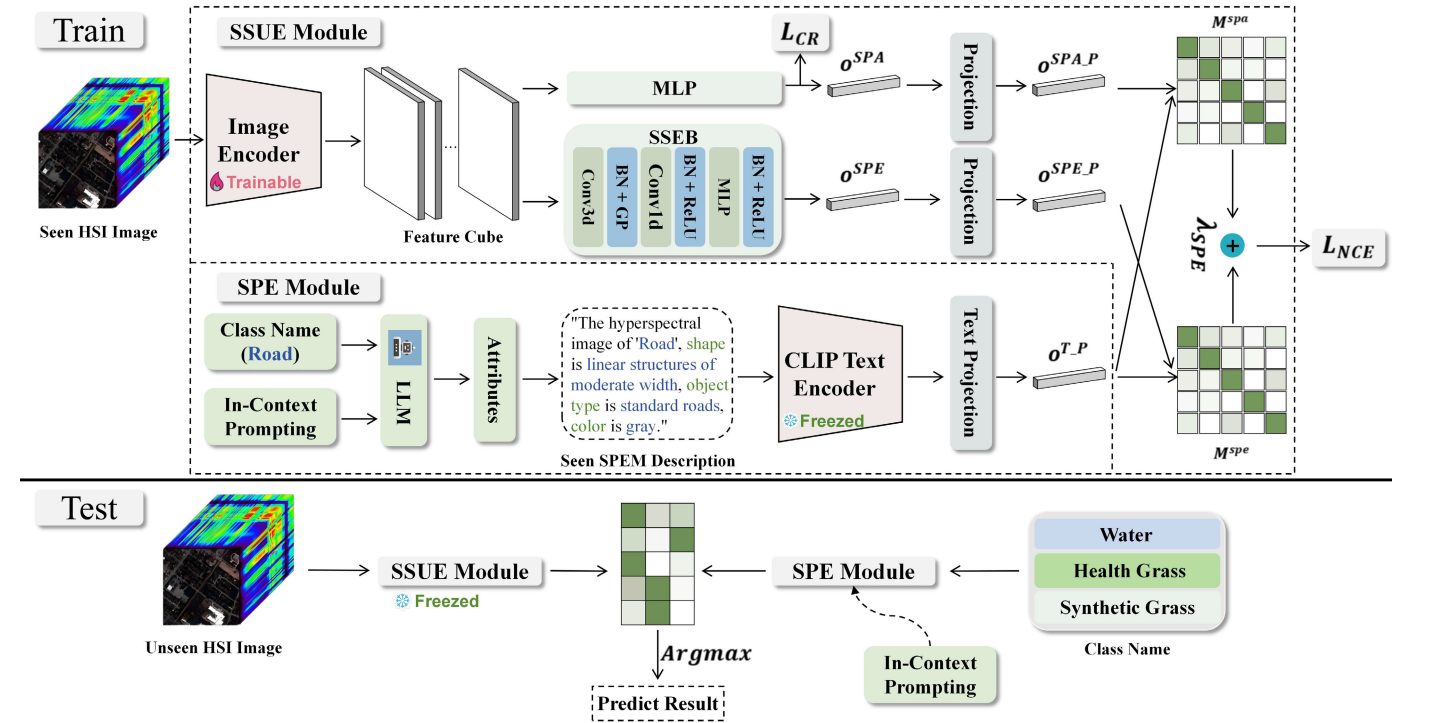


Fig. 2. Framework of the proposed SpectralZero. The SPE module encodes rich natural language descriptions into semantic features and projects them into both spatial and spectral embedding spaces. The SSUE module decouples image features into spatial and spectral components, which are then aligned with the corresponding semantic features. The similarity matrices from the two aligned spaces are fused and used to compute the loss during training. During testing, classification is performed by selecting the class index corresponding to the maximum value in the fused similarity matrix.

This significantly enhances the model's capacity for semantic modeling and recognition, providing critical support for zero-shot classification tasks.

3) An SSUE module is designed to improve the model's ability to extract spatial and spectral features under zero-shot conditions. The module consists of two independent branches: a spatial branch for capturing structural information of land covers, and a spectral branch for explicitly modeling the rich reflectance characteristics inherent in hyperspectral data. These features are then deeply fused to generate a unified embedding capable of jointly expressing spatial and spectral attributes, such as color and material properties, thereby significantly enhancing classification performance in zero-shot scenarios.

II. METHODOLOGY

This section presents the overall architecture of the proposed SpectralZero framework and provides a detailed description of its two core components: SPE module and SSUE module. In addition, Section II-D further elaborates on the implementation details of the model. The overall pipeline of SpectralZero is illustrated in Fig. 2.

A. Overview

Given an HSI dataset $D = \{\mathbf{I}, \mathbf{L}\}$, where $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ denotes an HSI, H and W represent the height and width of

the image, and C is the number of spectral bands. The set of labels is denoted as $L^N = \{l^1, l^2, \dots, l^N\}$. To facilitate model training and evaluation, we extract image patches of size $p \times p$ centered on specific pixels from image $\mathbf{I}$ in dataset D , and associate them with corresponding class labels to construct a subset $D^p = \{\mathbf{I}^p, \mathbf{L}\}$, where $\mathbf{I}^p \in \mathbb{R}^{p \times p \times C}$. According to the zero-shot classification setting, the subset D^p is further divided into seen classes subset $D_S^p = \{\mathbf{I}_S^p, \mathbf{L}_S^K\}$ and unseen classes subset $D_U^p = \{\mathbf{I}_U^p, \mathbf{L}_U^M\}$, where S and U denote the seen and unseen classes, K and M denote the number of seen and unseen classes, respectively. Each sample from seen classes subset contains an image patch $\mathbf{I}_S^p \in \mathbb{R}^{p \times p \times C}$, a label set $\mathbf{L}_S^K = \{l_S^1, l_S^2, \dots, l_S^K\} \in L^N$. The construction of the unseen class subset is similar to that of the seen classes, including image patches $\mathbf{I}_U^p \in \mathbb{R}^{p \times p \times C}$, the unseen label set $\mathbf{L}_U^M = \{l_U^1, l_U^2, \dots, l_U^M\} \in L^N$.

This article focuses on the strict zero-shot classification setting, where the label sets of seen and unseen classes are disjoint and satisfying

$$\mathbf{L}_S^K \cap \mathbf{L}_U^M = \emptyset. \quad (1)$$

The goal of strict zero-shot classification is to ensure that no unseen class samples participate in training. Under this constraint, only the seen class subset D_S^p is used as training data to learn a mapping function

$$f: \mathbf{I}_S^p \rightarrow \mathbf{I}_U^p \quad (2)$$

表 I 超参数定义

Symbol	Description
τ	Learnable temperature parameter in InfoNCE loss.
B	Batch size used in training.
λ_{sim}	Spectral similarity fusion coefficient, $\lambda_{\text{sim}} \in [0, 1]$.
λ_{loss}	Loss fusion coefficient, $\lambda_{\text{loss}} \in [0, 1]$.

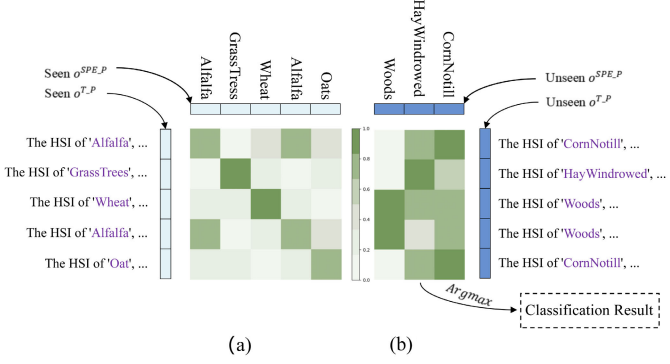


图 3. 所提模型在训练阶段和测试阶段的相似度矩阵。(a) 训练阶段。利用余弦相似度对两种不同模态的特征进行对齐, 生成相似度矩阵。相似度越高, 矩阵中对应的颜色越深。(b) 测试阶段。以与训练阶段相同的方式生成相似度矩阵, 并通过选取最大值对应的索引来获取预测结果。

使得该函数 f 能够对未见类别子集 $D_{\mathcal{U}}^P$ 实现精确分类。为清晰起见, 我们在术语表与表格I中总结了关键符号和超参数。

首先, 提出的 SpectralZero 框架包含两个主要阶段: 训练和测试。在训练阶段, 类别名称和来自可见类的对应高光谱影像数据 I_s^P 分别输入到 SPE 模块和 SSUE 模块。具体而言, SPE 模块利用预训练大语言模型的丰富知识和上下文推理能力, 在上下文提示的引导下, 自动生成语义增强的文本描述, 这些描述比原始类别名称提供更丰富的视觉-语义表达。随后, 这些语义增强的文本被输入到预训练的 CLIP 文本编码器 f_t 中, 以准确提取具有更强判别能力的文本语义特征 o^t 。同时, SSUE 模块首先从高光谱图像块中提取特征, 得到的低维特征被分离为光谱分支和空间分支。每个分支独立执行一维光谱特征提取和带有辅助分类的空间特征提取, 分别输出光谱特征 o^{spe} 和空间特征 o^{spa} 。接着, 提取的空间特征 o^{spa} 和光谱特征 o^{spe} 共同输入到图像投影中, 与文本语义特征 o^t 对齐, 从而构建两个跨模态嵌入空间 S^{spa} 和 S^{spe} 。这些视觉投影与文本语义投影 o^t 匹配, 生成两个相似性矩阵 M^{spa} 和 M^{spe} , 如图 3(a) 所示。最后, 采用加权融合策略将这两个相似性矩阵合并为一个统一相似度矩阵 M 。该统一相似度

矩阵随后通过 InfoNCE损失函数 [33] 进行优化。InfoNCE 损失的公式定义如下:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(M/\tau)}{\sum_{j=1}^B \exp(M_j/\tau)} \quad (3)$$

其中 τ 是一个可学习的温度参数, 用于控制训练过程中对比的平滑度, B 是批量大小。

在测试阶段, 来自未见类别数据集的语义增强文本和高光谱影像数据分别被输入到 SPE模块 和 SSUE模块 中, 以获取对应的文本特征和双分支光谱-空间特征。随后, 这三个特征进行对齐, 生成一个融合相似性矩阵, 如图 3(b) 所示。模型通过选取融合相似性矩阵每一行中最大值对应的索引来确定每个样本的预测类别, 从而完成高光谱图像的零样本分类任务。

B. SPE模块

本节介绍SPE模块, 该模块将类别标签扩展为带有尺度一致线索的属性级自然语言描述, 以增强跨模态对齐中的文本表达能力。具体而言, 以IP数据集为例, 我们首先为每个类别定义多个视觉属性维度, 包括形状(例如, 地表)、物体类型(例如, 作物类型)和颜色(代表光谱信息, 例如, 植被颜色), 并为每个属性提供相关的视觉描述。细粒度类别描述由预训练大语言模型通过固定提示模板生成, 且不进行任何后处理(例如, 清洗或过滤)和人工整理。对于每个数据集, 我们使用相同的提示模板, 以确保跨类别和数据集的语义生成一致性。然后, 采用类别前置策略, 将类别名称与每个视觉属性描述相结合, 如表格II所示, 生成一个全面SPE文本 $\mathbf{T}$, 该文本系统地传达了属性级类别语义, 提示模板和属性处理流程如图4所示。

为了增强语义提示的表达能力, 并确保生成的文本在捕获土地覆盖类别的视觉属性时兼具准确性与丰富性, SPE进一步利用预训练大语言模型, 基于类别名称和上下文提示, 自动生成覆盖多种视觉维度(如颜色、形状、材质和结构)的细粒度属性描述。重要的是, 可见类与不可见类的文本生成仅使用类别名称和数据集无关的领域知识, 且不访问目标数据集的任何像素、掩码、统计信息、地理元数据或图像衍生线索。因此, 来自不可见类的视觉或数据集特定信息不会泄露到训练中。这一策略不仅显著扩展了文本提示的语义表示能力, 还在语义嵌入空间中建立了更精确且更具判别性的类别区分。通过利用这种语义增强机制, 生成的文本描述能够在语义空间中实现与视觉特征更高质量的对其, 从而有效提升

TABLE I
HYPERPARAMETER DEFINITIONS

Symbol	Description
τ	Learnable temperature parameter in InfoNCE loss.
B	Batch size used in training.
λ_{sim}	Spectral similarity fusion coefficient, $\lambda_{\text{sim}} \in [0, 1]$.
λ_{loss}	Loss fusion coefficient, $\lambda_{\text{loss}} \in [0, 1]$.

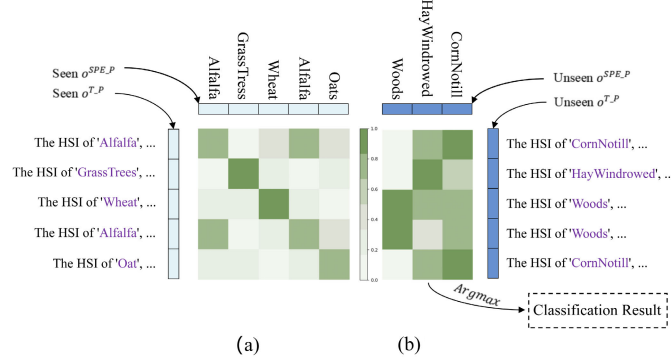


Fig. 3. Similarity matrix of the proposed model in both the training and testing stages. (a) Training stage. The features from the two different modalities are aligned using cosine similarity to generate a similarity matrix. Higher similarity corresponds to a darker color in the matrix. (b) Testing stage. The similarity matrix is generated in the same manner as in the training stage, and prediction results are obtained by selecting the indices corresponding to the maximum values.

such that the function f enables accurate classification on the unseen subset $D_{\mathcal{U}}^P$. For clarity, we summarize key notations and hyperparameters in Nomenclature and Table I.

First, the proposed SpectralZero framework consists of two main stages: training and testing. During the training stage, the class name and corresponding HSI data I_s^P from the seen classes are input into the *SPE module* and the *SSUE module*, respectively. Specifically, the SPE module leverages the rich knowledge and contextual reasoning capability of pretrained LLMs, guided by in-context prompts, to automatically generate semantically enhanced textual descriptions that provide richer visual-semantic expressions than the original class names. Subsequently, these semantic-enhanced texts are fed into the pretrained CLIP text encoder f_t to accurately extract textual semantic features o^t with stronger discriminative capability. Meanwhile, the SSUE module initially extracts features from the HSI patch, and the resulting low-dimensional features are separated into spectral and spatial branches. Each branch independently performs 1-D spectral feature extraction and spatial feature extraction with auxiliary classification, outputting spectral features o^{spe} and spatial features o^{spa} , respectively. Subsequently, the extracted spatial features o^{spa} and spectral features o^{spe} are jointly input into the image projection to align with textual semantic features o^t , thereby constructing two cross-modal embedding spaces, S^{spa} and S^{spe} . These visual projections are matched with text semantic projections o^t to generate two similarity matrices, M^{spa} and M^{spe} , as depicted in Fig. 3(a). Finally, a weighted fusion strategy is applied to combine these two similarity matrices into a unified similarity matrix M . This unified similarity

matrix is then optimized via the InfoNCE loss function [33]. The formula of the InfoNCE loss is defined as follows:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(M/\tau)}{\sum_{j=1}^B \exp(M_j/\tau)} \quad (3)$$

where τ is a learnable temperature parameter that controls the smoothness of the contrast during training and B is the batch size.

During the testing phase, the semantic-enhanced texts and HSI data from the unseen class dataset are, respectively, fed into the SPE module and the SSUE module to obtain the corresponding text features and dual-branch spectral-spatial features. These three features are then aligned to generate a fused similarity matrix, as illustrated in Fig. 3(b). The model determines the predicted class of each sample by selecting the index of the maximum value in each row of the fused similarity matrix, thereby completing the zero-shot classification task for HSI.

B. SPE Module

This section introduces the SPE module, which expands class labels into attribute-level natural language descriptions with scale-consistent cues to enhance textual expressiveness for cross-modal alignment. Specifically, taking the IP dataset, we first define multiple visual attribute dimensions for every class, including shape (e.g., land surface), object type (e.g., crop type), and color (represent spectral information, e.g., vegetation color), and provide relevant visual descriptions for each attribute. Fine-grained class descriptions are generated by a pretrained LLM using a fixed prompt template and no post-processing (e.g., cleaning or filtering) and no manual curation are applied. For each dataset, we use the same prompt template to ensure consistent semantic generation across classes and datasets. Then, using a class-prepend strategy, the class name is combined with each visual attribute description which are listed in Table II to generate a comprehensive SPE text $\mathbf{T}$ that systematically conveys attribute-level class semantics, the prompting template and the attribute processing flow are shown in Fig. 4.

To enhance the expressive ability of semantic prompts and ensure the accuracy and richness of the generated texts in capturing visual attributes of land-cover categories, SPE further utilizes a pretrained LLM to automatically generate fine-grained attribute descriptions covering multiple visual dimensions, such as color, shape, material, and structure, based on class names and in-context prompts. Importantly, the text generation for seen and unseen classes uses only the class names and dataset-agnostic domain knowledge, and does not access any pixels, masks, statistics, geographic metadata, or image-derived cues from the target datasets. Thus, no visual or dataset-specific information from unseen classes is leaked into training. This strategy not only significantly expands the semantic representation capacity of textual prompts but also establishes more precise and discriminative distinctions among classes within the semantic embedding space. By leveraging this semantic enrichment mechanism, the generated textual descriptions achieve higher quality alignment with visual features in the semantic space, thereby effectively enhancing the

表II视觉属性描述 (以IP数据集为例)

Class	Attributes	Description	Text
CornNotill	crop type land surface other attributes	corn flat green	The hyperspectral image of ‘CornNotill’, crop type is corn, land surface is flat, vegetation color is green.
SoybeanMintill	crop type land surface other attributes	soybean undulating mintill	The hyperspectral image of ‘SoybeanMintill’, crop type is soybean, land surface is undulating, vegetation color is green.
CornMintill	crop type land surface other attributes	corn undulating mintill	The hyperspectral image of ‘CornMintill’, crop type is corn, land surface is undulating, tillage is mintill.

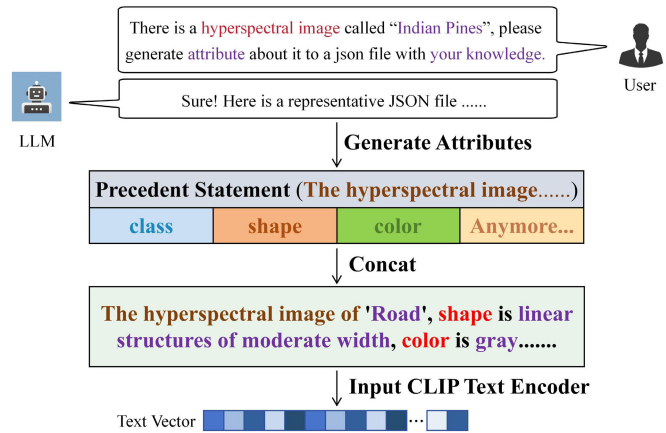


图4. 语义增强文本生成 workflow。我们首先向大语言模型提供包含关键术语的提示, 要求其生成属性级描述, 如形状、颜色和植被类别。随后, 将这些属性与预定义模板拼接, 形成增强语义提示, 再输入CLIP文本编码器, 以获取可靠的文本特征表示。

模型对图像内容的理解能力及其在零样本场景中的判别性能。图5 进一步展示了模型如何通过语言信息将知识从可见类迁移到不可见类的过程。

为了精确捕捉SPE文本中蕴含的视觉-语义信息, 我们进一步引入冻结的CLIP文本编码器来编码增强后的描述。CLIP模型在大规模图像-文本对上训练, 具备强大的文本理解与语义表示能力。通过利用预训练的CLIP文本编码器, 该模型能够提取准确且具有判别性的文本语义特征。随后, 提取的文本特征通过映射头投影到SSUE中的两个独立跨模态嵌入空间。这些嵌入分别与图像的空间特征和光谱特征进行对齐, 从而实现特征对齐和跨模态分类。

C. 光谱-空间联合特征提取模块

尽管初始的3D-CNN在学习光谱-空间联合表示方面表现出强大的能力, 但其

输出特征在空间信息和光谱响应方面本质上是纠缠在一起的。当这种纠缠的表征被直接投影到统一的跨模态嵌入空间时, 空间上下文的变化可能会干扰光谱语义, 导致语义错位并降低跨模态对齐的可辨别性。同时, 这种耦合使得在对齐过程中难以明确且可控地调整空间与光谱信息的相对贡献。因此, 我们提出了SSUE模块来解耦初始特征, 并分别提取空间信息和光谱响应特征, 为后续的对齐和融合提供更可控的表征基础。该模块包含四个子组件: 一个初始图像编码器、一个一维光谱提取分支、一个空间提取与分类分支, 以及一个双分支对齐相似度加权融合机制。

初始图像编码器采用自 [30], 包含多个三维卷积块, 旨在从输入图像中提取光谱-空间联合特征 o^i 。然而, 这种表示方式固在地混合了空间和光谱信息, 对图像光谱特征的精确提取产生不利影响。因此, 我们在SSUE模块中引入了单维光谱提取分支 (SSEB), 旨在减少空间图像特征的干扰, 增强模型的光谱提取能力。具体而言, 该光谱提取分支首先通过三维卷积神经网络层对初始特征 o^i (由初始图像编码器提取) 进行降维, 生成维度为 $C' \times H' \times W'$ 的中间特征。随后, 应用全局池化将这些中间特征沿空间维度聚合成一维光谱特征向量 o^{spe} 。接着, 该光谱向量通过一维卷积神经网络层进行特征细化, 以提取局部光谱特征, 再通过额外的批归一化层捕获非线性光谱特征。最后, 所得的光谱特征 o^{spe} 被投影到空间-文本跨模态联合嵌入空间 S^{spe} 中, 用于后续的跨模态对齐和相似度加权融合。

同时, 为了有效提取和分类空间特征, 由初始图像编码器获得的光谱-空间联合特征被展平为

TABLE II
VISUAL ATTRIBUTE DESCRIPTIONS (IP DATASET FOR EXAMPLE)

Class	Attributes	Description	Text
CornNotill	crop type land surface other attributes	corn flat green	The hyperspectral image of ‘CornNotill’, crop type is corn, land surface is flat, vegetation color is green.
SoybeanMintill	crop type land surface other attributes	soybean undulating mintill	The hyperspectral image of ‘SoybeanMintill’, crop type is soybean, land surface is undulating, vegetation color is green.
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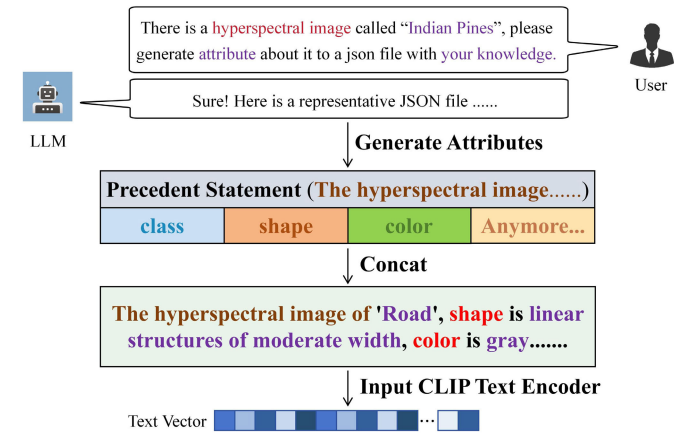


Fig. 4. Semantic enhancement text generation workflow. We first provide the LLM with a prompt containing key terms and ask it to generate attribute-level descriptions, such as shape, color, and vegetation category. These attributes are then concatenated with a predefined template to form an enriched semantic prompt, which is subsequently fed into the CLIP text encoder to obtain reliable textual feature representations.

model’s understanding of image content and its discriminative performance in zero-shot scenarios. Fig. 5 further illustrates the process by which the model transfers knowledge from seen to unseen classes through linguistic information.

To precisely capture the visual-semantic information embedded in the SPE text, we further introduce the CLIP text encoder which is frozen to encode the enriched descriptions. The CLIP model, trained on large-scale image–text pairs, possesses strong capabilities in textual understanding and semantic representation. By leveraging the pretrained CLIP text encoder, the model can extract accurate and discriminative textual semantic features. Subsequently, the extracted textual features are projected by a mapping head into two separate cross-modal embedding spaces within SSUE. These embeddings are then aligned with the spatial and spectral features of the image, respectively, to enable feature alignment and cross-modal classification.

C. Spectral–Spatial Union Feature Extraction Module

Although the preliminary 3D-CNN demonstrates strong capability in learning spectral–spatial joint representations, its

output features are inherently entangled in terms of spatial information and spectral responses. When such entangled representations are directly projected into a unified cross-modal embedding space, variations in spatial context may interfere with spectral semantics, resulting in semantic misalignment and reduced discriminability of cross-modal alignment. Meanwhile, this coupling makes it difficult to explicitly and controllably adjust the relative contributions of spatial and spectral information during alignment. Therefore, we propose the SSUE module to decouple the initial features and separately extract spatial information and spectral response characteristics, providing a more controllable representational basis for subsequent alignment and fusion. The proposed module comprises four subcomponents: a preliminary image encoder, a 1-D spectral extraction branch, a spatial extraction and classification branch, and a dual-branch alignment similarity-weighted fusion mechanism.

The preliminary image encoder is adopted from [30], comprising multiple 3-D convolutional blocks designed to extract spectral–spatial joint features o^i from input images. However, this representation inherently mixes spatial and spectral information, adversely affecting the precise extraction of image spectral features. Thus, we introduce a single-dimensional spectral extraction branch (SSEB) within the SSUE module, aiming to reduce interference from spatial image features and enhance the spectral extraction capability of the model. Specifically, this spectral extraction branch initially performs dimensionality reduction on preliminary features o^i (extracted by the preliminary image encoder) through a 3D-CNN layer, generating intermediate features with dimensions $C' \times H' \times W'$. Subsequently, global pooling is applied to aggregate these intermediate features along spatial dimensions into a 1-D spectral feature vector o^{spe} . Next, the spectral vector undergoes feature refinement through 1D-CNN layers to extract local spectral features, followed by additional BN layers for capturing nonlinear spectral characteristics. Finally, the resultant spectral features o^{spe} are projected into the spatial–text cross-modal union embedding space S^{spe} for subsequent cross-modal alignment and similarity-weighted fusion.

Concurrently, to effectively extract and classify spatial features, the spectral–spatial joint features o^i obtained by the

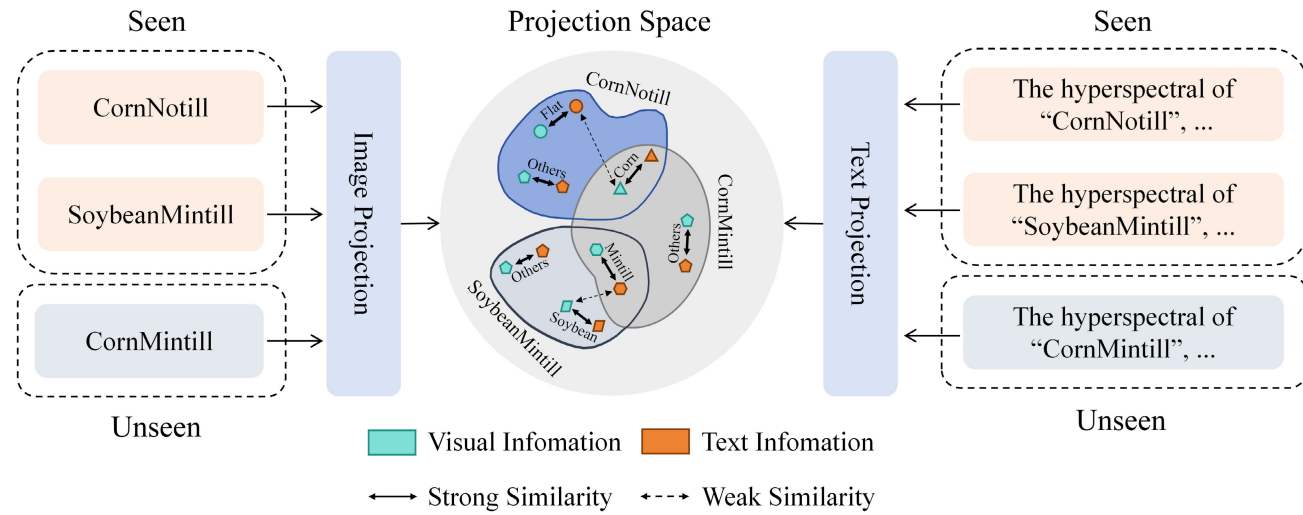


图 5. 投影空间的可视化，其中视觉特征与文本特征实现了对齐。视觉特征与文本特征被映射到一个共享投影空间中。在此空间内，绿色形状代表视觉信息，橙色形状代表文本信息，并使用不同的几何形状来区分不同类别。映射后的可见类与不可见类共享视觉与文本语义信息，这些信息充当了知识传递锚点。同时，在对齐约束下，同一类别的视觉与文本特征保持高度相似性，而不同类别的视觉特征则被推离与之对应的文本特征。

一维特征向量。该向量随后通过两条并行路径同时处理，具体如下。

- 1) 将其输入到一个全连接分类器层，得到针对可见类的类别置信度向量 p^{cls} 。交叉熵损失 $\mathcal{L}_{CE}$ 与真实标签一起用于监督训练，以优化初始图像编码器。
- 2) 该向量被投影到空间-文本跨模态联合嵌入空间 S^{spa} ，从而支持后续的跨模态特征对齐与相似度加权融合。

在双分支相似性加权融合机制中，模型同时提取图像和文本特征，最终获得由光谱、空间和文本语义特征构成的多模态表征。光谱特征和文本语义特征通过投影变换被投影到一个统一的光谱-文本跨模态对齐空间中，

$$\begin{aligned} o^{spe_p} &= \mathbf{W}_{spe} \cdot o^{spe} \\ o^{t_p} &= \mathbf{W}_t \cdot o^t \end{aligned} \quad (4)$$

其中， o^{spe_p} 和 o^{t_p} 分别代表投影后的光谱特征和文本特征， $\mathbf{W}_{spe}$ 和 $\mathbf{W}_t$ 是光谱特征和文本特征对应的投影矩阵。类似地，空间特征和文本特征被投影到空间-文本跨模态对齐空间中，如下所示：

$$\begin{aligned} o^{spa_p} &= \mathbf{W}_{spa} \cdot o^{spa} \\ o^{t_p} &= \mathbf{W}_t \cdot o^t \end{aligned} \quad (5)$$

其中 o^{spa_p} 表示投影后的空间特征， $\mathbf{W}_{spa}$ 代表空间特征投影矩阵。这两个跨模态空间分别实现了高光谱图像与文本在光谱和空间维度上的高级别对齐。在完成跨模态特征对齐后，

计算得到光谱-文本相似性矩阵 $\mathbf{M}^{spe}$ 和空间-文本相似性矩阵 $\mathbf{M}^{spa}$ 。为进一步融合光谱与空间信息，我们采用比例加权策略将这两个矩阵合并为一个融合相似性矩阵 $\mathbf{M}$

$$\mathbf{M} = (1 - \lambda_{sim}) \cdot \text{sim}(o^{spa_p}, o^{t_p}) + \lambda_{sim} \cdot \text{sim}(o^{spe_p}, o^{t_p}). \quad (6)$$

由此产生的融合相似性矩阵 $\mathbf{M}$ 随后被用于计算 InfoNCE [33] 损失，以在训练过程中优化模型。这里， λ_{sim} 是一个可调的超参数，称为光谱相似性融合系数，其取值范围为 $\lambda_{sim} \in [0, 1]$ 。sim($\cdot$) 表示用于衡量两个向量之间相似度的余弦相似度运算。在上式中， o^{spa_p} 和 o^{spe_p} 分别表示投影后的空间特征和光谱特征， o^{t_p} 表示投影后的文本特征。最后，InfoNCE 损失与交叉熵损失通过加权求和相结合，得到总损失函数

$$\mathcal{L}_{total} = \lambda_{loss} \cdot \mathcal{L}_{CE} + (1 - \lambda_{loss}) \cdot \mathcal{L}_{InfoNCE} \quad (7)$$

其中 λ_{loss} 是一个控制损失权重的超参数，称为损失融合系数，其值范围为 $\lambda_{loss} \in [0, 1]$ 。该参数在训练过程中动态平衡监督分类学习和对比学习。

D. 实现

为清晰说明所提模型的训练实现过程，算法 1 给出了由 SPE 模块和 SSUE 模块组成的模型完整训练流程的伪代码。在 SPE 模块中，模型对图像进行初步编码，并分别提取光谱特征与空间特征。光谱特征通过独立分支进行编码和投影，而空间特征既用于生成辅助分类信号，也用于空间-语义对齐。

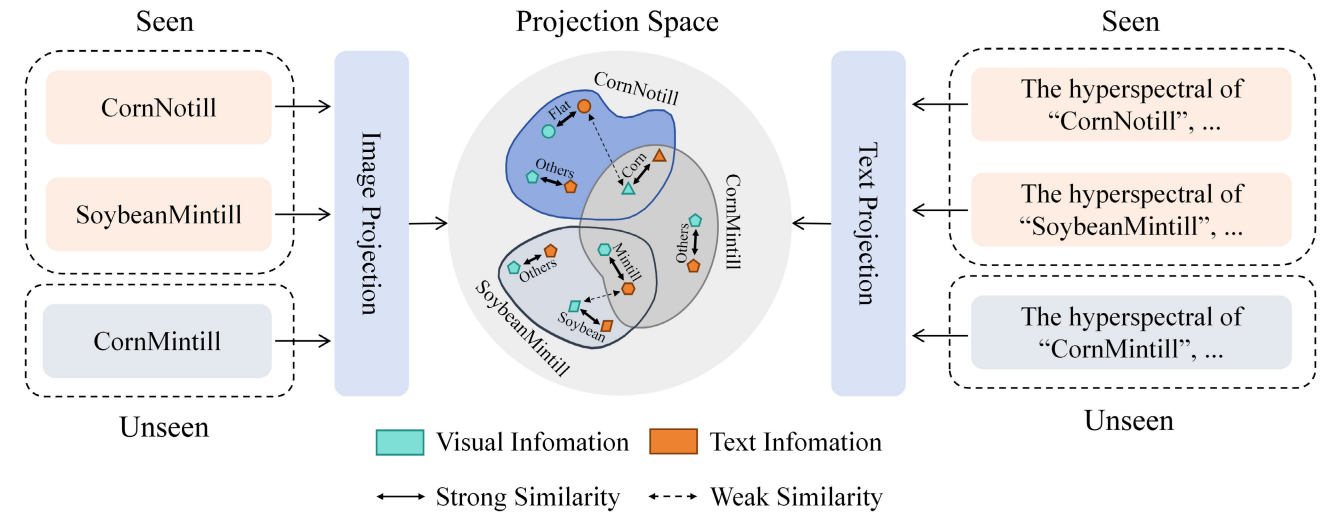


Fig. 5. Visualization of the projection space where visual and text features are aligned. Visual features and textual features are mapped into a shared projection space. Within this space, green shapes represent visual information and orange shapes denote textual information, with different geometric shapes used to distinguish different categories. The mapped seen and unseen classes share visual and textual semantic information, which serves as anchors for knowledge transfer. Meanwhile, under the alignment constraints, visual and textual features from the same category maintain high similarity, whereas visual features are pushed away from textual features of different categories.

preliminary image encoder are flattened into a 1-D feature vector o^i . This vector is then simultaneously processed through two parallel pathways as follows.

- 1) It is input to a fully connected classifier layer, yielding a class confidence vector p^{cls} for seen classes. Cross-entropy loss $\mathcal{L}_{CE}$ is employed with ground-truth labels for supervised training to facilitate optimization of the preliminary image encoder.
- 2) The vector is projected into the spatial-text cross-modal union embedding space S^{spa} , thus, supporting subsequent cross-modal feature alignment and similarity-weighted fusion.

In the dual-branch similarity-weighted fusion mechanism, the model simultaneously extracts image and textual features, ultimately obtaining multimodal representations composed of spectral, spatial, and textual semantic features. The spectral features and textual semantic features are projected into a unified spectral-text cross-modal alignment space via projection transformations

$$\begin{aligned} o^{spe_p} &= \mathbf{W}_{spe} \cdot o^{spe} \\ o^{t_p} &= \mathbf{W}_t \cdot o^t \end{aligned} \quad (4)$$

where o^{spe_p} and o^{t_p} represent the projected spectral and text features, respectively, and $\mathbf{W}_{spe}$ and $\mathbf{W}_t$ are the corresponding projection matrices for spectral and textual features. Similarly, spatial features and text features are projected into the spatial-text cross-modal alignment space as follows:

$$\begin{aligned} o^{spa_p} &= \mathbf{W}_{spa} \cdot o^{spa} \\ o^{t_p} &= \mathbf{W}_t \cdot o^t \end{aligned} \quad (5)$$

where o^{spa_p} denotes the projected spatial features, and $\mathbf{W}_{spa}$ represents the spatial feature projection matrix. These two cross-modal spaces separately enable high-level alignment between HSI and text in the spectral and spatial dimensions. After completing cross-modal feature alignment, the

spectral-text similarity matrix $\mathbf{M}^{spe}$ and spatial-text similarity matrix $\mathbf{M}^{spa}$ are computed. To further fuse spectral and spatial information, we employ a proportional weighting strategy to combine these two matrices into a fused similarity matrix $\mathbf{M}$

$$\mathbf{M} = (1 - \lambda_{sim}) \cdot \text{sim}(o^{spa_p}, o^{t_p}) + \lambda_{sim} \cdot \text{sim}(o^{spe_p}, o^{t_p}). \quad (6)$$

The resulting fused similarity matrix $\mathbf{M}$ is subsequently utilized to compute the InfoNCE [33] loss for model optimization during training. Here, λ_{sim} is a tunable hyperparameter termed the spectral similarity fusion coefficient, with a value range of $\lambda_{sim} \in [0, 1]$. sim($\cdot$) denotes the cosine similarity operation used to measure similarity between two vectors. In the above, o^{spa_p} and o^{spe_p} denote the spatial and spectral features after projection, respectively, and o^{t_p} denotes the projected textual features. Finally, the InfoNCE loss is combined with the cross-entropy loss through weighted summation, yielding the overall loss function

$$\mathcal{L}_{total} = \lambda_{loss} \cdot \mathcal{L}_{CE} + (1 - \lambda_{loss}) \cdot \mathcal{L}_{InfoNCE} \quad (7)$$

where λ_{loss} is a hyperparameter controlling the loss weight, termed the loss fusion coefficient, with a value range of $\lambda_{loss} \in [0, 1]$. This parameter dynamically balances supervised classification learning and contrastive learning during the training process.

D. Implementation

To clearly illustrate the training implementation process of the proposed model, Algorithm 1 presents the pseudocode for the complete training procedure of the model composed of the SPE module and SSUE module. In the SPE module, the model performs preliminary encoding of the image and separately extracts spectral and spatial features. The spectral features are encoded and projected through an independent branch, while the spatial features are utilized both for generating auxiliary classification signals and for spatial-semantic alignment.

算法1 配备SPE模块和SSUE模块的SpectralZero

输入: 训练数据集 $D_S^P = \{I_S^P, L_S^M\}$
输出: 训练好的模型

- 1: 对于所有 样本对 (i_s, l_s) 在 D_S^P 中执行
- 2: // SSUE模块
- 3: // 提取初始视觉特征
- 4: $o^i \leftarrow$ 视觉编码器(i_s)
- 5: // 提取光谱特征
- 6: $o^{spe} \leftarrow$ 单光谱提取(o^i)
- 7: // 提取空间特征与分类信号
- 8: $(o^{spa}, p^{cls}) \leftarrow$ 空间特征提取(o^i)
- 9: // 辅助分类信号
- 10: $\mathcal{L}_{CE} \leftarrow$ 交叉熵损失(p^{cls}, l_s)
- 11: $o^{spa}_{\mathcal{P}} \leftarrow$ 空间投影头(o^i)
- 12: $o^{spe}_{\mathcal{P}} \leftarrow$ 光谱投影头(o^{spe})
- 13:
- 14: // SPE模块
- 15: $t_s \leftarrow$ generateEnrichmentText(名称,属性)
- 16: $o^t \leftarrow$ CLIPTextEncoder(t_s)
- 17: $o^t_{\mathcal{P}} \leftarrow$ textProjectionHead(o^t) // 文本投影
- 18:
- 19: // 对齐 (空间-文本 和 频谱-文本)
- 20: $M^{spa} \leftarrow$ 余弦相似度($o^t_{\mathcal{P}}, o^{spa}_{\mathcal{P}}$) -
- 21: $M^{spe} \leftarrow$ 余弦相似度($o^t_{\mathcal{P}}, o^{spe}_{\mathcal{P}}$) -
- 22:
- 23: // 相似度融合
- 24: $M \leftarrow (1 - \lambda_{\text{sim}}) M^{spa} + \lambda_{\text{sim}} M^{spe}$
- 25: $\mathcal{L}_{\text{InfoNCE}} \leftarrow$ InfoNCELoss(M)
- 26: // 融合损失
- 27: $\mathcal{L}_{\text{total}} \leftarrow \lambda_{\text{loss}} \mathcal{L}_{CE} + (1 - \lambda_{\text{loss}}) \mathcal{L}_{\text{InfoNCE}}$
- 28:
- 29: // 更新
- 30: $\mathcal{L}_{\text{total}}.\text{backward}()$
- 31: 更新(model.parameter)
- 32: 结束循环

SSUE模块使用冻结的CLIP文本编码器提取语义特征，并将其映射到统一语义空间，以便与图像特征进行双路径对齐。在此基础上，构建两个相似性矩阵并通过加权组合进行融合，模型使用InfoNCE [33] 损失函数优化图像-文本对齐能力，同时融合辅助分类信号进行端到端训练。该实现过程为后续的实验设置和多模型比较提供了统一的训练基础。具体的超参数配置将在第III节进一步详述。

III. 实验

A. 实验设置

1) 数据集: 我们在三个公开数据集上进行了实验，以验证我们的模型在高光谱图像分类中的有效性: Indian Pines (IP) 数据集、Houston 2013 (Houston) 数据集和WHU-Hi-LongKou (WHHL)[34], [35] 数据集。数据集配置总结在表III中，我们从每个可见类中随机选取20%的样本进行训练

，而所有不可见类的样本则用于测试。为确保公平和全面的评估，我们在不同折数中对不可见类采用随机旋转，并在所有实验中固定由此产生的划分，以便每个类别恰好被作为不可见类处理一次。

a) *Indian Pines*: IP数据集由AVIRIS传感器于1992年采集。它由一幅大小为 145×145 的高光谱图像组成，包含224个光谱波段，覆盖了从0.4到2.5 μm 的电磁波谱。由于部分波段位于水吸收区域，按照惯例去除了24个波段 (104–108、150–163和220)，最终得到200个有效波段。该图像主要覆盖农业和植被区域，约三分之二为农作物相关土地覆盖，其余三分之一主要为森林或混合植被。共包含16个类别，如“Woods”和“Grass Trees”。我们选择了13个类别作为训练集，其余三个类别作为测试集。假彩色图像和类别标签如图6所示。

b) *Houston 2013*: 休斯顿数据集是我们研究中使用的基准数据集之一。它由ITRES CASI-1500传感器于2013年采集，覆盖休斯顿大学校园及周边区域，包含144个光谱波段，范围从0.38到1.05 μm 。图像尺寸为 349×1905 像素，空间分辨率为2.5米。该数据集主要代表典型的的城市土地覆盖类型，包含15个标注类别，包括“高速公路”和“商业区”。我们选择了12个类别作为训练集，其余3个类别保留用于测试集。假彩色图像和类别标签如图7所示。

c) *WHU-Hi-LongKou* [34], [35]: WHHL数据集于2018年在中国湖北省龙口镇采集，使用安装在无人机 (UAV) 平台上的Headwall Nano-Hyperspec高光谱成像系统，飞行高度约500米。采集的图像经过了辐射定标和几何校正，确保了良好的物理一致性和空间准确性。WHHL图像包含270个光谱波段，覆盖范围从0.4到1.0 μm ，尺寸为 550×400 像素，空间分辨率为0.463米。该数据集主要覆盖农业区域，标注了9种典型的土地覆盖类别，如“水稻”和“棉花”。我们选择了6个类别作为训练集，其余3个类别用于测试。假彩色图像和类别标签如图8所示。

2) 评估指标: 为全面评估模型在零样本分类上的性能，采用了四种标准指标: 总体精度 (OA)、平均精度 (AA)、卡帕系数、每类精度和变异系数 (CV)。总体精度通过计算正确预测样本与未见类别样本总数的比率来量化模型的泛化能力。每类精度通过衡量每个类别内正确预测的比例来评估类别特定的识别能力，而平均精度则通过对所有类别的每类精度取平均值，反映整体类别间的平衡性。卡帕系数进一步评估预测标签与真实标签之间的统计一致性，

Algorithm 1 SpectralZero With SPE Module and SSUE Module

Input: Training datasets $D_S^P = \{I_S^P, L_S^M\}$
Output: Trained model

- 1: **for all** sample-pair (i_s, l_s) in D_S^P **do**
- 2: // SSUE module
- 3: // Extract initial visual feature
- 4: $o^i \leftarrow$ visualEncoder(i_s)
- 5: // Extract spectral feature
- 6: $o^{spe} \leftarrow$ singleSpectralExtract(o^i)
- 7: // Extract spatial feature and classification signal
- 8: $(o^{spa}, p^{cls}) \leftarrow$ spatialFeatureExtract(o^i)
- 9: // Auxiliary classification signal
- 10: $\mathcal{L}_{CE} \leftarrow$ crossEntropyLoss(p^{cls}, l_s)
- 11: $o^{spa}_{\mathcal{P}} \leftarrow$ spatialProjectionHead(o^i)
- 12: $o^{spe}_{\mathcal{P}} \leftarrow$ spectralProjectionHead(o^{spe})
- 13:
- 14: // SPE module
- 15: $t_s \leftarrow$ generateEnrichmentText($Name, Attributes$)
- 16: $o^t \leftarrow$ CLIPTextEncoder(t_s)
- 17: $o^t_{\mathcal{P}} \leftarrow$ textProjectionHead(o^t) // Text projection
- 18:
- 19: // Alignment (spatial-text and spectral-text)
- 20: $M^{spa} \leftarrow$ cosineSimilarity($o^t_{\mathcal{P}}, o^{spa}_{\mathcal{P}}$)
- 21: $M^{spe} \leftarrow$ cosineSimilarity($o^t_{\mathcal{P}}, o^{spe}_{\mathcal{P}}$)
- 22:
- 23: // Similarity fusion
- 24: $M \leftarrow (1 - \lambda_{\text{sim}}) M^{spa} + \lambda_{\text{sim}} M^{spe}$
- 25: $\mathcal{L}_{\text{InfoNCE}} \leftarrow$ InfoNCELoss(M)
- 26: // Fusion loss
- 27: $\mathcal{L}_{\text{total}} \leftarrow \lambda_{\text{loss}} \mathcal{L}_{CE} + (1 - \lambda_{\text{loss}}) \mathcal{L}_{\text{InfoNCE}}$
- 28:
- 29: // Update
- 30: $\mathcal{L}_{\text{total}}.\text{backward}()$
- 31: update(model.parameter)
- 32: **end for**

The SSUE module extracts semantic features using a frozen CLIP text encoder and maps them into a unified semantic space for dual-path alignment with the image features. Based on this, two similarity matrices are constructed and fused via weighted combination, and the model is optimized for image-text alignment capability using the InfoNCE [33] loss, while the auxiliary classification signal is simultaneously fused for end-to-end training. This implementation process provides a unified training foundation for subsequent experimental settings and multimodel comparisons. The specific hyperparameter configurations will be further detailed in Section III.

III. EXPERIMENTS

A. Experimental Settings

1) *Datasets*: We conducted experiments on three public datasets to validate the effectiveness of our model capacity in HSI classification: the Indian Pines (IP) dataset, the Houston 2013 (Houston) dataset, and the WHU-Hi-LongKou (WHHL) [34], [35] dataset. Dataset configurations are summarized in Table III, we randomly select 20% of the samples from each

seen class for training, while all samples from unseen classes are used for testing. To ensure a fair and comprehensive evaluation, we adopt random rotation for the unseen classes across folds and then keep the resulting splits fixed for all experiments, such that each class is treated as unseen exactly once.

a) *Indian Pines*: The IP dataset was collected by the AVIRIS sensor in 1992. It consists of an HSI of size 145×145 with 224 spectral bands, covering the electromagnetic spectrum from 0.4 to 2.5 μm . Since some bands are located in water absorption regions, 24 bands (104–108, 150–163, and 220) were removed following standard practice, resulting in 200 effective bands. The image mainly covers agricultural and vegetation areas, with approximately two-thirds consisting of crop-related land covers and the remaining one-third predominantly forests or mixed vegetation. A total of 16 classes are included, such as “Woods” and “Grass Trees.” We selected 13 classes for the training set, while the remaining three classes were designated as the test set. The false-color image and class labels are shown in Fig. 6.

b) *Houston 2013*: The Houston dataset is one of the benchmark datasets utilized in our study. It was acquired by the ITRES CASI-1500 sensor in 2013 over the University of Houston campus and its surrounding areas, comprising 144 spectral bands ranging from 0.38 to 1.05 μm . The image dimension is 349×1905 pixels with a spatial resolution of 2.5 m. This dataset primarily represents typical urban land-cover types, with 15 annotated classes, including “Highway” and “Commercial.” We selected 12 classes as the training set, and the remaining three classes were reserved for the test set. The false-color image and class labels are illustrated in Fig. 7.

c) *WHU-Hi-LongKou* [34], [35]: The WHHL dataset was acquired in 2018 over Longkou Town, Hubei Province, China, using a Headwall Nano-Hyperspec hyperspectral imaging system mounted on an uncrewed aerial vehicle (UAV) platform at approximately 500-m altitude. The collected images underwent radiometric calibration and geometric correction, ensuring good physical consistency and spatial accuracy. The WHHL image consists of 270 spectral bands covering a range from 0.4 to 1.0 μm , with dimensions of 550×400 pixels and a spatial resolution of 0.463 m. This dataset predominantly covers agricultural areas, annotated with 9 typical land-cover classes such as “Rice” and “Cotton.” We selected six classes for the training set, while the remaining three classes were used for testing. The false-color image and class labels are depicted in Fig. 8.

2) *Evaluation Metrics*: To comprehensively evaluate the model’s performance on zero-shot classification, four standard metrics are employed: overall accuracy (OA), average accuracy (AA), Kappa coefficient, per-class accuracy, and coefficient of variation (CV). OA quantifies the model’s generalization by computing the ratio of correctly predicted samples to the total number of unseen class samples. Per-class accuracy assesses class-specific recognition by measuring the proportion of correct predictions within each class, while AA reflects overall class-wise balance by averaging per-class accuracy across all categories. The Kappa coefficient further evaluates the statistical consistency between predicted and true labels,

表IIIIP数据集、休斯顿和WHHL数据集的设置（在20%的可见类上训练；每折三个不可见类用于测试，非重叠旋转）

Group	Class	IP		Class	Houston		Class	WHHL	
		Total	Train(Seen)		Total	Train(Seen)		Total	Train(Seen)
1	CornNotill	1428	285	Healthy Grass	1251	250	Corn	34511	6902
	HayWindrowed	478	95	Synthetic Grass	697	139	Cotton	8374	1674
	Woods	1265	253	Water	325	65	Water	67056	13411
2	GrassTrees	730	146	Commercial	1244	248	Sesame	3031	606
	BuildGraTreeDri	386	77	Railway	1235	247	Broad-Leaf Soybean	63212	12642
	SoybeanMintill	2455	491	Tree	1244	248	Rice	11854	2370
3	GrassPasMow	28	15	Tennis Court	428	85	Narrow-Leaf Soybean	4151	830
	StoneSteTow	93	18	Running Track	660	132	Mixed Weed	5229	1424
	CornMintill	830	166	Parking Lot 1	1233	246	Roads And Houses	7124	1045
4	Corn	237	47	Parking Lot 2	469	93			
	Alfalfa	46	15	Residential	1268	253			
	GrassPasture	483	96	Road	1252	250			
5	Wheat	205	41	Stressed Grass	1254	250			
	Oats	20	15	Soil	1242	248			
	SoybeanNotill	972	194	Highway	1227	245			
6	SoybeanClean	593	118						
Total		10249	2072	Total	15029	2999	Total	204542	40904

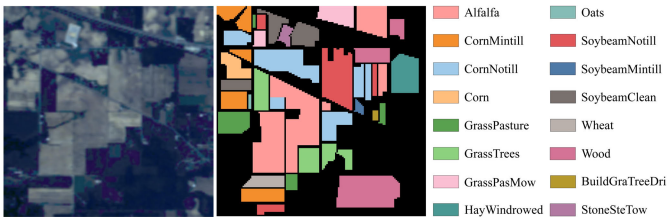


图6. IP数据集的假彩色图像及其对应的真值图。

有效排除了随机一致性的影响，增强了性能评估的鲁棒性。为进一步证明模型的稳定性，我们利用变异系数评估潜在的每类偏差。变异系数定义为每类精度的标准差与均值的比值，其表达式为

$$CV = \frac{\sigma}{\mu} \quad (8)$$

其中 σ 和 μ 分别表示每类精度的标准差和均值。较低的变异系数意味着更均衡的每类性能，表明更好的模型稳定性。

3) 训练细节：SpectralZero基于PyTorch框架实现。实验在一台配备Intel¹ 至强¹ Platinum 8352V CPU（2.10 GHz）、24 GB英伟达GeForce RTX 4090 GPU和90 GB系统内存的机器上进行。我们使用GPT-4作为大语言模型来生成SPE文本。

4) 参数设置：在本节中，我们描述了若干对模型性能影响相对较小的超参数设置，包括批量大小、学习

¹注册商标。

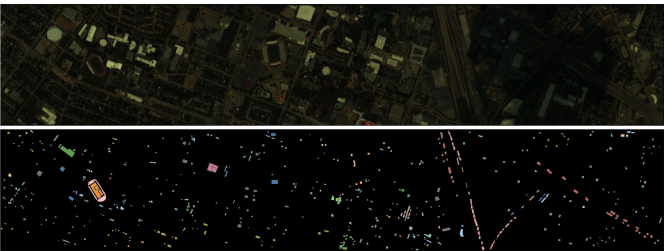


图 7. 休斯顿数据集的假彩色图像及其对应的真值图。

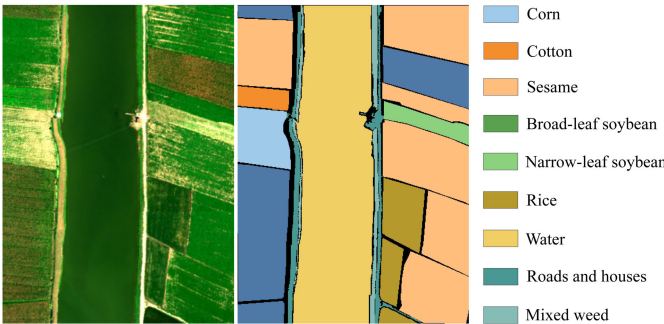


图 8. WHHL数据集的假彩色图像及其对应的真值图。

率、训练轮数、优化器选择以及对比损失中使用的温度系数。为平衡训练效率与内存限制，所有数据集的批量大小统一设置为1024，训练样本比例设置为20%。基于各数据集特性，我们采用不同的初始学习率：IP数据集为0.0001，

TABLE III
SETTINGS FOR THE IP, HOUSTON, AND WHHL DATASETS (TRAIN ON 20% OF SEEN CLASSES; THREE UNSEEN CLASSES PER FOLD FOR TESTING, NONOVERLAPPING ROTATIONS)

Group	Class	IP		Class	Houston		Class	WHHL	
		Total	Train(Seen)		Total	Train(Seen)		Total	Train(Seen)
1	CornNotill	1428	285	Healthy Grass	1251	250	Corn	34511	6902
	HayWindrowed	478	95	Synthetic Grass	697	139	Cotton	8374	1674
	Woods	1265	253	Water	325	65	Water	67056	13411
2	GrassTrees	730	146	Commercial	1244	248	Sesame	3031	606
	BuildGraTreeDri	386	77	Railway	1235	247	Broad-Leaf Soybean	63212	12642
	SoybeanMintill	2455	491	Tree	1244	248	Rice	11854	2370
3	GrassPasMow	28	15	Tennis Court	428	85	Narrow-Leaf Soybean	4151	830
	StoneSteTow	93	18	Running Track	660	132	Mixed Weed	5229	1424
	CornMintill	830	166	Parking Lot 1	1233	246	Roads And Houses	7124	1045
4	Corn	237	47	Parking Lot 2	469	93			
	Alfalfa	46	15	Residential	1268	253			
	GrassPasture	483	96	Road	1252	250			
5	Wheat	205	41	Stressed Grass	1254	250			
	Oats	20	15	Soil	1242	248			
	SoybeanNotill	972	194	Highway	1227	245			
6	SoybeanClean	593	118						
Total		10249	2072	Total	15029	2999	Total	204542	40904

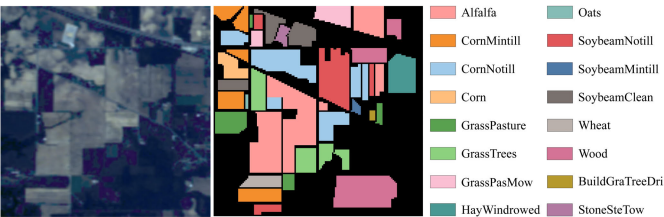


Fig. 6. False-color image and its corresponding ground-truth map of the IP dataset.

effectively accounting for chance agreement and enhancing the robustness of performance evaluation. To further demonstrate the stability of our model, we evaluate potential per-class bias using the CV. CV is defined as the ratio between the standard deviation and the mean of the per-class accuracies, which can be expressed as

$$CV = \frac{\sigma}{\mu} \quad (8)$$

where σ and μ denote the standard deviation and mean of the per-class accuracies, respectively. A lower CV implies more balanced per-class performance, indicating better model stability.

3) *Training Details*: SpectralZero is implemented based on the PyTorch framework. The experiments are conducted on a machine equipped with an Intel¹ Xeon¹ Platinum 8352V CPU@2.10 GHz and a 24-GB NVIDIA GeForce RTX 4090 GPU, with 90 GB of system memory. We use GPT-4 as the LLM to generate SPE texts.

4) *Parameter Setting*: In this section, we describe several hyperparameter settings that have a relatively minor impact on model performance, including batch size, learning

¹Registered trademark.

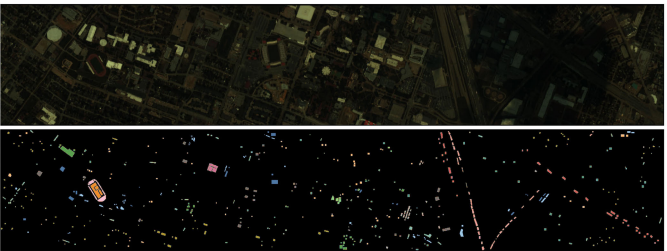


图 7. False-color image and its corresponding ground-truth map of the Houston dataset.

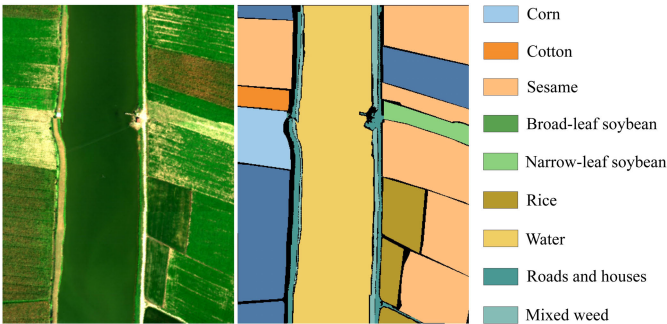


Fig. 8. False-color image and its corresponding ground-truth map of the WHHL dataset.

rate, number of training epochs, optimizer selection, and the temperature coefficient used in contrastive loss. To balance training efficiency with memory constraints, the batch size is uniformly set to 1024 and the train sample is set to 20% for all datasets. Based on the characteristics of each dataset, we adopt different initial learning rates: 0.0001 for IP, 0.0002 for

表IVIP数据集上的零样本分类实验结果

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	CornNotill HayWindrowed Woods	74.26	86.73	66.07	72.88	0.00	99.84
		30.44	14.01	65.90	0.22	0.00	94.39
		67.17	96.01	29.47	98.34	100.00	67.99
Group 2	GrassTrees BuildGraTreeDri SoybeanMintill	99.49	6.33	26.88	0.00	2.40	99.67
		8.93	0.00	0.00	0.00	0.00	61.60
		0.00	98.72	90.40	100.00	0.00	97.45
Group 3	GrassPasMow StoneSteTow CornMintill	100.00	69.23	0.00	0.00	0.00	100.00
		0.00	0.00	45.33	100.00	0.00	0.00
		0.00	98.60	64.07	0.00	0.00	97.13
Group 4	Corn Alfalfa GrassPasture	2.10	0.00	0.00	100.00	98.93	100.00
		0.00	51.61	0.00	3.23	0.00	0.00
		99.74	93.02	100.00	0.00	67.37	93.58
Group 5	Wheat Oats SoybeanNotill	0.00	0.00	86.59	0.00	58.08	99.66
		100.00	92.07	0.00	100.00	0.00	100.00
		44.22	74.68	53.98	0.00	0.00	92.08
Group 6	SoybeanClean	79.79	98.32	60.00	100.00	99.90	99.85
Results	OA (%)	42.60	74.00	57.53	30.31	29.56	97.34
	AA (%)	44.13	48.81	43.04	42.16	26.67	81.45
	$\kappa \times 100$	16.20	30.06	13.69	8.81	17.74	87.78

休斯顿数据集为0.0002，WHHL数据集为0.00002。关于训练轮数，IP数据集最大轮数设为100，休斯顿数据集为150，WHHL数据集为50。IP和休斯顿数据集每10轮评估一次模型，WHHL数据集每5轮评估一次。由于最终性能选取所有评估点中的最优值，因此轮数的微小调整对最终结果影响甚微。我们统一采用AdamW优化器 [36] 跨所有数据集，该优化器将权重衰减与基于梯度的参数更新解耦，从而提升模型泛化能力。对比损失中的温度系数在训练过程中保持可学习状态，初始化为：IP和休斯顿数据集0.1，WHHL数据集0.17。为进一步增强模型鲁棒性，训练期间对所有数据集应用翻转增强和随机噪声增强。在图块构建方面，为提升特征提取质量，我们将IP、休斯顿和WHHL数据集的图块尺寸分别设置为15、11和13。

B. 性能分析

在我们的对比实验中，不仅包含了传统的零样本分类模型，还纳入了几种专门针对高光谱图像零样本分类设计的代表性方法。所选基准方法包括：f-CLSWGAN [22]，APN [37]，CE-GZSL [38],ACMR [39],和SPECIAL [32]。

1) f-CLSWGAN [22]作为零样本分类领域的代表性方法，f-CLSWGAN联合训练一个特征生成器和一个判别器，并引入辅助分类损失以增强生成特征的判别性。最终，该方法利用

生成器为不可见类合成特征，并将其与可见类样本结合，用于训练下游分类器，从而实现对不可见类别的识别。

2) APN [37] 通过将局部图像特征与类别属性特征对齐来实现零样本分类，同时利用全局图像特征信息进行最终判别。

3) CE-GZSL [38]提出了一种混合型广义零样本分类框架，该框架将特征生成与判别性嵌入相结合。通过将真实样本和合成样本映射到对比嵌入空间，并结合类别级和实例级监督，模型提升了分类性能。鉴于原始实现支持零样本分类(ZSC)设置，我们在严格零样本分类配置下对该模型进行训练和评估，以便进行对比。

4) ACMR [39] 通过构建视觉变分自编码器和文本变分自编码器对数据分布进行建模，并引入IEM模块以防止潜在空间坍塌，同时增强视觉与文本模态之间的跨模态信息融合。

5) SPECIAL [32] 模型的详细描述见第I节。作为针对高光谱图像零样本分类的最新方法，它在性能和方法论设计上均展现出显著优势，因此成为有力的对比基准。

为确保公平比较，所有方法均使用相同的SPE文本作为输入进行评价；除非另有说明，其余超参数和实现细节均遵循各自论文中报告的最佳性能设置。

TABLE IV
ZERO-SHOT CLASSIFICATION RESULTS OF EXPERIMENTS ON THE IP DATASET

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	CornNotill HayWindrowed Woods	74.26	86.73	66.07	72.88	0.00	99.84
		30.44	14.01	65.90	0.22	0.00	94.39
		67.17	96.01	29.47	98.34	100.00	67.99
Group 2	GrassTrees BuildGraTreeDri SoybeanMintill	99.49	6.33	26.88	0.00	2.40	99.67
		8.93	0.00	0.00	0.00	0.00	61.60
		0.00	98.72	90.40	100.00	0.00	97.45
Group 3	GrassPasMow StoneSteTow CornMintill	100.00	69.23	0.00	0.00	0.00	100.00
		0.00	0.00	45.33	100.00	0.00	0.00
		0.00	98.60	64.07	0.00	0.00	97.13
Group 4	Corn Alfalfa GrassPasture	2.10	0.00	0.00	100.00	98.93	100.00
		0.00	51.61	0.00	3.23	0.00	0.00
		99.74	93.02	100.00	0.00	67.37	93.58
Group 5	Wheat Oats SoybeanNotill	0.00	0.00	86.59	0.00	58.08	99.66
		100.00	92.07	0.00	100.00	0.00	100.00
		44.22	74.68	53.98	0.00	0.00	92.08
Group 6	SoybeanClean	79.79	98.32	60.00	100.00	99.90	99.85
Results	OA (%)	42.60	74.00	57.53	30.31	29.56	97.34
	AA (%)	44.13	48.81	43.04	42.16	26.67	81.45
	$\kappa \times 100$	16.20	30.06	13.69	8.81	17.74	87.78

Houston, and 0.00002 for WHHL. Regarding training epochs, the maximum number of epochs is set to 100 for IP, 150 for Houston, and 50 for WHHL. The model is evaluated every 10 epochs for IP and Houston, and every 5 epochs for WHHL. Since the final performance is selected as the best among all evaluation points, minor adjustments to the number of epochs have a negligible influence on the final results. We employ the AdamW optimizer [36] across all datasets, which decouples weight decay from the gradient-based parameter updates, thereby improving model generalization. The temperature coefficient in the contrastive loss remains learnable throughout training and is initialized to 0.1 for IP and Houston and to 0.17 for WHHL. To further improve model robustness, we apply flip augmentation and random noise augmentation to all datasets during training. For patch construction, in order to enhance feature extraction quality, we set the patch size to 15, 11, and 13 for IP, Houston, and WHHL, respectively.

B. Performance Analysis

In our comparative experiments, we include not only traditional zero-shot classification models but also several representative methods specifically designed for HSI zero-shot classification. The selected baselines include: f-CLSWGAN [22], APN [37], CE-GZSL [38], ACMR [39], and SPECIAL [32].

1) f-CLSWGAN [22] as a representative approach in the field of zero-shot classification, f-CLSWGAN jointly trains a feature generator and a discriminator, incorporating an auxiliary classification loss to enhance the discriminability of the generated features. Ultimately, this method synthesizes features for unseen classes using

the generator and combines them with seen class samples to train a downstream classifier, thereby enabling recognition of unseen categories.

2) APN [37] achieves zero-shot classification by aligning local image features with class attribute features, while also leveraging global image feature information for final discrimination.

3) CE-GZSL [38] proposes a hybrid generalized zero-shot classification framework that integrates feature generation with discriminative embedding. By mapping both real and synthesized samples into a contrastive embedding space and combining class-level and instance-level supervision, the model enhances classification performance. Given that the original implementation supports the ZSC setting, we train and evaluate this model under the strict zero-shot classification configuration for comparison.

4) ACMR [39] models the data distribution by constructing both visual VAEs and textual VAEs, and introduces an IEM module to prevent latent space collapse and to enhance cross-modal information integration between visual and textual modalities.

5) The SPECIAL [32] model has been detailed in Section I. As the most recently proposed method for HSI zero-shot classification, it demonstrates significant advantages in both performance and methodological design, making it a strong baseline for comparison.

To ensure a fair comparison, all methods are evaluated with the same SPE text as input; unless otherwise noted, all remaining hyperparameters and implementation details follow the best-performing settings reported in their respective papers.

表V休斯顿数据集上的零样本分类实验结果

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	Healthy Grass Synthetic Grass Water	93.15	94.20	83.16	99.43	93.57	100.00
		19.71	92.94	54.34	3.07	0.00	81.54
		75.23	8.54	33.31	0.00	91.91	97.63
Group 2	Commercial Railway Tree	0.00	14.26	85.14	99.70	0.30	36.53
		98.99	15.89	0.00	0.00	0.00	69.60
		91.27	89.23	21.39	0.00	0.00	85.78
Group 3	Tennis Court Running Track Parking Lot 1	1.17	13.99	58.31	0.29	0.00	74.06
		1.70	1.33	0.00	85.23	0.00	48.94
		100.00	97.57	74.77	37.99	72.29	80.86
Group 4	Parking Lot 2 Residential Road	12.77	1.06	80.05	55.59	0.00	57.67
		15.07	92.41	45.62	5.32	0.00	62.06
		79.24	27.05	0.00	86.73	1.90	61.54
Group 5	Stressed Grass Soil Highway	100.00	99.50	100.00	70.62	0.00	99.80
		0.00	98.09	0.00	0.00	0.00	0.20
		0.00	0.00	0.00	64.36	100.00	100.00
Results	OA (%)	52.09	58.64	43.78	45.05	27.49	71.36
	AA (%)	45.88	50.35	42.40	40.55	24.00	70.35
	$\kappa \times 100$	22.53	30.11	13.91	11.39	15.12	55.79

表VIWHHL数据集上的零样本分类实验结果

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	Corn Cotton Water	33.93	60.04	39.57	79.13	96.60	99.26
		37.79	63.76	59.41	3.74	49.97	85.40
		98.94	97.78	98.93	98.81	99.98	100.00
Group 2	Sesame Broad-Leaf Soybean Rice	10.14	0.00	0.00	1.79	0.00	93.70
		87.72	100.00	1.79	23.42	0.00	1.47
		87.12	0.00	99.23	4.03	5.93	89.50
Group 3	Narrow-Leaf Soybean Mixed Weed Roads And Houses	74.57	0.00	77.60	0.18	1.06	95.50
		32.13	22.11	75.94	44.12	0.00	18.76
		32.22	87.84	11.46	25.23	69.08	26.00
Results	OA (%)	68.54	69.75	47.47	43.33	42.02	76.46
	AA (%)	54.95	47.95	51.55	31.16	35.66	67.74
	$\kappa \times 100$	44.74	25.15	28.14	13.53	36.75	59.48

表IV–VI 展示了所提方法与几种现有零样本分类模型在三个高光谱图像数据集（IP、休斯顿和WHHL）上的实验结果。评估指标包括总体精度、平均精度、每类分类结果以及Kappa系数。

如表IV–VI所示，SpectralZero在所有三个数据集上均实现了最佳的零样本分类整体性能。在基线方法中，APN在多个类别上表现稳定，在三个数据集上均取得了特别高的总体精度和平均精度。相比之下，CE-GZSL、ACMR和SPECIAL在大多数类别的分类性能上表现出一定的局限性。f-CLSWGAN模型的竞争力较弱。

在训练样本较少的场景中（例如IP数据集和休斯顿数据集），这可能是由于其高度依赖数据规模；然而，在WHHL数据集上，由于可用的训练样本更多，其性能有所提升。总体而言，所提出的方法在IP数据集、休斯顿数据集和WHHL数据集上的总体精度（OA）分别比第二优模型高出23.34%（APN）、12.72%（APN）和6.71%（APN）。就平均精度（AA）而言，我们的方法在IP数据集、休斯顿数据集和WHHL数据集上分别比各自第二优模型高出32.64%（APN）、20.00%（APN）和12.79%（f-CLSWGAN），充分证明了我们的方法在零样本泛化能力方面的显著优势。

TABLE V
ZERO-SHOT CLASSIFICATION RESULTS OF EXPERIMENTS ON THE HOUSTON DATASET

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	Healthy Grass Synthetic Grass Water	93.15	94.20	83.16	99.43	93.57	100.00
		19.71	92.94	54.34	3.07	0.00	81.54
		75.23	8.54	33.31	0.00	91.91	97.63
Group 2	Commercial Railway Tree	0.00	14.26	85.14	99.70	0.30	36.53
		98.99	15.89	0.00	0.00	0.00	69.60
		91.27	89.23	21.39	0.00	0.00	85.78
Group 3	Tennis Court Running Track Parking Lot 1	1.17	13.99	58.31	0.29	0.00	74.06
		1.70	1.33	0.00	85.23	0.00	48.94
		100.00	97.57	74.77	37.99	72.29	80.86
Group 4	Parking Lot 2 Residential Road	12.77	1.06	80.05	55.59	0.00	57.67
		15.07	92.41	45.62	5.32	0.00	62.06
		79.24	27.05	0.00	86.73	1.90	61.54
Group 5	Stressed Grass Soil Highway	100.00	99.50	100.00	70.62	0.00	99.80
		0.00	98.09	0.00	0.00	0.00	0.20
		0.00	0.00	0.00	64.36	100.00	100.00
Results	OA (%)	52.09	58.64	43.78	45.05	27.49	71.36
	AA (%)	45.88	50.35	42.40	40.55	24.00	70.35
	$\kappa \times 100$	22.53	30.11	13.91	11.39	15.12	55.79

TABLE VI
ZERO-SHOT CLASSIFICATION RESULTS OF EXPERIMENTS ON THE WHHL DATASET

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	Corn Cotton Water	33.93	60.04	39.57	79.13	96.60	99.26
		37.79	63.76	59.41	3.74	49.97	85.40
		98.94	97.78	98.93	98.81	99.98	100.00
Group 2	Sesame Broad-Leaf Soybean Rice	10.14	0.00	0.00	1.79	0.00	93.70
		87.72	100.00	1.79	23.42	0.00	1.47
		87.12	0.00	99.23	4.03	5.93	89.50
Group 3	Narrow-Leaf Soybean Mixed Weed Roads And Houses	74.57	0.00	77.60	0.18	1.06	95.50
		32.13	22.11	75.94	44.12	0.00	18.76
		32.22	87.84	11.46	25.23	69.08	26.00
Results	OA (%)	68.54	69.75	47.47	43.33	42.02	76.46
	AA (%)	54.95	47.95	51.55	31.16	35.66	67.74
	$\kappa \times 100$	44.74	25.15	28.14	13.53	36.75	59.48

Tables IV–VI present the experimental results of the proposed method and several existing zero-shot classification models on three HSI datasets: IP, Houston, and WHHL. The evaluation metrics include OA, AA, per-class classification results, and the Kappa coefficient.

As shown in Tables IV–VI, SpectralZero achieves the best zero-shot classification overall performance on all three datasets. Among the baseline methods, APN demonstrates stable performance across multiple categories, achieving particularly high OA and AA on the three datasets. On the contrary, CE-GZSL, ACMR, and SPECIAL exhibit certain limitations in classification performance for most categories. The f-CLSWGAN model performs less competitively

in scenarios with fewer training samples (such as IP and Houston), which may be due to its high dependency on data scale; however, its performance improves on the WHHL dataset, where more training samples are available. Overall, the proposed method outperforms the second-best models by 23.34% (APN), 12.72% (APN), and 6.71% (APN) in OA on the IP, Houston, and WHHL datasets, respectively. For the AA, our method achieves improvements of 32.64% (APN), 20.00% (APN), and 12.79% (f-CLSWGAN) over the respective second-best models on the IP, Houston, and WHHL datasets, fully demonstrating the significant advantage of our method in zero-shot generalization capability.

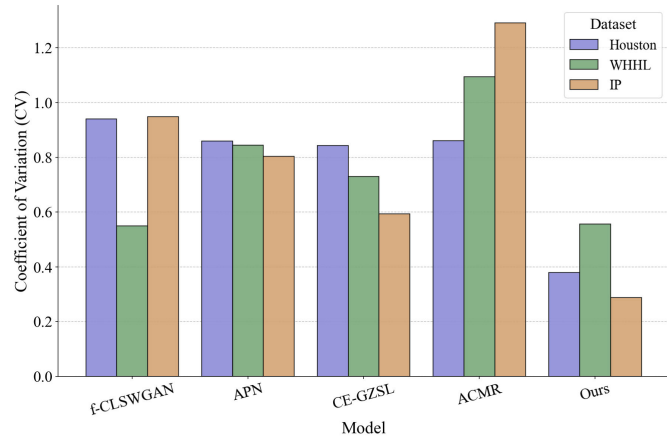


图9. 三个数据集上各模型的变异系数。y轴数值越低表示鲁棒性越强。

此外, 基于表IV–VI中的各类别结果, 我们分析了跨不同数据集的类别级性能。在IP数据集上, 我们的方法对大多数类别实现了较高的识别准确性, 例如CornNotilt为99.84%, Corn为100.00%。然而, 少数类别表现出明显的失败, 例如第3组中StoneSteTow的准确性为0.00%, 第4组中Alfalfa的准确性为0.00%。这一现象主要归因于这些类别的有效样本数量相对有限 (少于20个), 以及Alfalfa和Stone Steel Tower在细粒度光谱-空间表示中存在显著的类内变异, 这可能会使类原型和投影特征分布不稳定, 从而削弱跨模态对齐的辨别能力。

在休斯顿数据集上, 性能在类别难度上表现出更明显的分歧。一方面, 我们的方法在具有相对一致的材料响应和清晰空间结构的类别上表现强劲, 例如健康草地和高速公路, 两者均达到100.00%, 这表明这些类别更有可能形成稳定且可对齐的跨模态语义锚点。另一方面, 在复杂的城市场景下, 一些类别仍然具有挑战性, 甚至出现极端失败——例如, 商业区仅达到36.53%, 土壤降至0.20%。这根本上是由于这些类别更强的类内异质性和更严重的背景混合效应。以土壤为例, 它通常与裸地、建筑区域或路边材料交织在一起, 导致混合光谱响应并与其他硬表面类别重叠, 这大大增加了区分难度。

在WHHL数据集上, 各类别的结果也显示出明显的类别差异。我们的方法在水体和Corn等类别上取得了较高的准确性, 而阔叶大豆、混合杂草以及道路和房屋则表现出明显的失败。这种行为与其空间分布模式密切相关: 这些困难类别通常呈现为狭长带状区域或边界过渡区域, 并且与相邻的土地覆盖类型紧密相邻, 使得块级表示更容易受到周围背景干扰的影响, 进而引发误分类并加剧类间混淆。

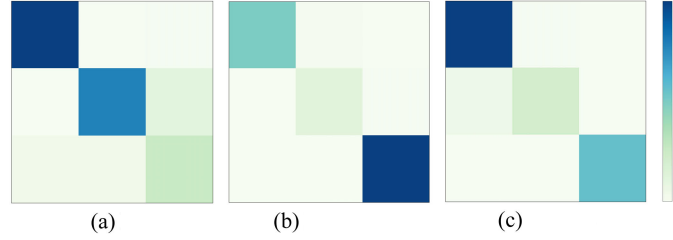


图 10. 所提模型在 (a) IP、(b) 休斯顿和 (c) WHHL数据集上的混淆矩阵。行表示真实类别, 列表示预测类别。矩阵以计数形式显示: 对角线元素表示每个类别中正确分类的样本数, 而非对角线元素表示被误分类到其他类别的样本数, 反映了类间混淆。颜色越深表示数值越高。该混淆矩阵基于每个数据集在三类旋转协议中的第一次划分计算得出, 旨在直观展示典型的混淆模式。

为了严格评估跨数据集的类别级偏差和鲁棒性, 我们采用变异系数 (其定义见(8)) 作为稳定性指标, 以量化每类的分类准确率, 结果如图9所示。在大多数设置下, 我们的方法在所有三个数据集上都获得了较低的变异系数值, 表明其类别性能更加稳定且偏差更小。

总之, 所提模型通过从混合特征中独立提取光谱信息并构建语义对齐路径, 辅以细粒度文本描述, 增强了模型对类别间光谱差异的感知。对于空间结构高度相似的类别 (例如 “小麦” 和 “燕麦”), 对光谱信息的建模能力能够实现更精确的区分; 对于依赖颜色和光谱描述类别 (例如 “HayWindrowed”), 语义提示文本进一步强化了模型对光谱语义特征的理解与表征, 从而提升了整体分类性能。

为进一步分析所提模型对每个类别的分类能力, 图 10 展示了该方法在三个数据集上的混淆矩阵。从图中可以看出, 大部分分类结果高度集中于主对角线, 表明该模型能准确区分各个数据集中的不同类别, 并取得了优异的总体分类性能。同时, 混淆矩阵非对角线区域的误分类数量相对较少, 这证明模型在处理具有相似光谱特征或空间结构的类别时, 仍保持良好的判别能力。这些结果验证了该方法在复杂场景下的有效性和鲁棒性, 并进一步确认了该模型在零样本设置下的泛化能力。

C. 模型分析

1) 消融研究: 为了进一步评估在相似性融合加权中融入光谱信息和语义提示增强文本的有效性, 我们设计了以下几个消融实验场景。

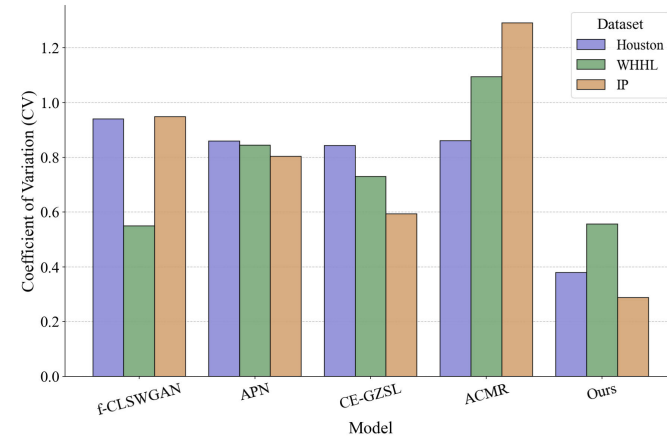


Fig. 9. CV across models on three datasets. Lower values on the y-axis indicate greater robustness.

Furthermore, based on the per-class results in Tables IV–VI, we analyze the class-level performance across different datasets. On the IP dataset, our method achieves high recognition accuracy for most classes, e.g., 99.84% for CornNotilt and 100.00% for Corn. Nevertheless, a few classes exhibit notable failures, such as 0.00% accuracy for StoneSteTow in Group 3 and 0.00% for Alfalfa in Group 4. This phenomenon is mainly attributed to the relatively limited number of effective samples for these classes (fewer than 20), as well as pronounced intraclass variations in fine-grained spectral–spatial representations for Alfalfa and Stone Steel Tower, which can destabilize class prototypes and the projected feature distribution, thereby weakening the discriminability of cross-modal alignment.

On the Houston dataset, the performance shows a more pronounced divergence in class difficulty. On the one hand, our method performs strongly on classes with relatively consistent material responses and clear spatial structures, such as healthy grass and highway, both reaching 100.00%, indicating that these classes are more likely to form stable and alignable cross-modal semantic anchors. On the other hand, under complex urban scenarios, some classes remain challenging and even suffer extreme failures—for example, commercial achieves only 36.53%, and soil drops to 0.20%. This is fundamentally due to stronger intraclass heterogeneity and more severe background mixing effects for these classes. Taking Soil as an example, it is often interwoven with bare land, construction areas, or road-edge materials, leading to mixed spectral responses and overlap with other hard-surface classes, which substantially increases the discrimination difficulty.

On the WHHL dataset, the per-class results also reveal clear class-wise disparities. Our method attains high accuracy on classes such as water and corn, while broad-leaf soybean, mixed weed, and roads and house exhibit evident failures. This behavior is closely related to their spatial distribution patterns: these difficult classes often appear as narrow strip-like regions or boundary-transition areas and are tightly adjacent to neighboring land-cover types, making patch-level representations more susceptible to surrounding background interference, which in turn induces misclassifications and exacerbates interclass confusion.

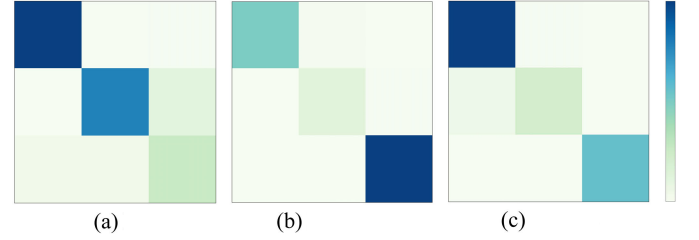


Fig. 10. Confusion matrices of the proposed model on (a) IP, (b) Houston, and (c) WHHL datasets. The rows denote the ground-truth classes and the columns denote the predicted classes. The matrices are shown in count form: diagonal entries indicate the number of correctly classified samples for each class, while off-diagonal entries indicate the number of samples misclassified into other classes, reflecting interclass confusion. Darker colors indicate higher values. This confusion matrix is computed from the first split in the three-class rotation protocol for each dataset and is provided to visually illustrate typical confusion patterns.

To rigorously assess class-level bias and robustness across datasets, we adopt the CV, which has been defined as in (8), as a stability metric to quantify per-class classification accuracy, the results are shown in Fig. 9. Across most settings, our method attains lower CV values on all three datasets, indicating more stable and less biased per-class performance.

In summary, the proposed model enhances the perception of spectral differences between categories by independently extracting spectral information from mixed features and constructing a semantic alignment pathway, complemented by fine-grained textual descriptions. For categories with highly similar spatial structures (e.g., “wheat” and “oats”), the ability to model spectral information enables more accurate differentiation; for categories that rely on color and spectral descriptions (e.g., “HayWindrowed”), semantic prompt texts further strengthen the model’s understanding and representation of spectral-semantic features, thereby improving overall classification performance.

To further analyze the classification capability of the proposed model for each category, Fig. 10 presents the confusion matrices of our method on the three datasets. As can be observed from the figure, most classification results are highly concentrated along the main diagonal, indicating that the model can accurately distinguish between different classes in each dataset and achieves strong overall classification performance. Meanwhile, the number of misclassifications in the off-diagonal regions of the confusion matrices is relatively small, demonstrating that the model still maintains good discriminative ability when dealing with classes that have similar spectral characteristics or spatial structures. These results validate the effectiveness and robustness of our approach in complex scenarios, and further confirm the generalization capability of the model under zero-shot settings.

C. Model Analysis

1) *Ablation Study*: To further evaluate the effectiveness of incorporating spectral information and semantic prompt-enhanced texts in the similarity fusion weighting, we design several ablation experiment scenarios as follows.

表 VII在零样本设置下, 针对 IP 数据集、休斯顿数据集和 WHHL 数据集所有类别的消融研究结果 (%)

Method	Setting	IP			Houston			WHHL		
		OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Baseline	Setting 1	93.35	76.90	75.79	67.63	66.99	49.78	73.60	63.24	54.08
w/o SPE module	Setting 2	94.41	80.76	80.05	68.04	65.97	49.52	74.86	66.94	55.70
w/o SSUE module	Setting 3	96.71	81.02	85.03	68.11	68.32	50.81	74.37	63.38	55.24
Full	Setting 4	97.34	81.45	87.78	71.36	70.35	55.79	76.46	67.74	59.48

1) 设置1: 仅使用基线模型, 不包含光谱-文本跨模态对齐空间, 也不使用SSEB进行特征提取。2) 设置2 (*SPE*模块消融): 将SPE文本替换为原始类别名称。该设置用于评估丰富的文本提示在表征视觉特征方面的贡献。3) 设置3 (*SSUE*模块消融): 将相似性融合系数 λ_{sim} 设为0。该设置用于分析SSUE模块中光谱特征提取能力对模型性能的贡献。4) 设置4: 完整的 SpectralZero模型。

消融实验的结果如表 VII所示。基于这些结果, 可以得出以下结论。

1) 将设置 4 与设置 2 进行比较, 我们观察到在 IP 数据集、休斯顿数据集和 WHHL 数据集上, 总体精度分别提升了 2.93%、3.32% 和 1.60%, 平均精度分别提升了 0.69%、4.38% 和 0.80%。这些结果表明, 语义提示增强文本在提升模型表达土地覆盖类型视觉特征的能力方面发挥着重要作用, 并有效促进了图像与文本特征之间的语义对齐。

2) 将设置 4 与设置 3 进行比较, 在 IP 数据集、休斯顿数据集和 WHHL 数据集上, 总体精度分别提升了 0.63%、3.32% 和 2.09%, 平均精度分别提升了 0.43 %、2.03% 和 4.36%。总体而言, 这些结果验证了 SSUE 模块在光谱维度上构建语义对齐的有效性, 这在具有高光谱可区分性的数据集上尤为明显。

3) 与设置1相比, 设置4在所有三个数据集上均实现了显著的性能提升, 其中总体精度 (OA) 在IP、休斯顿 (Houston) 和WHHL上分别提升了3.99%、3.73%和2.86%, 平均精度 (AA) 分别提升了4.55% (IP)、3.36% (休斯顿) 和 4.50% (WHHL)。这些结果充分证明了所提出的双分支对齐相似性加权融合策略能够有效增强模型在高光谱图像零样本分类任务中的判别能力与泛化性能。

2) 超参数分析: 本节分析了几个对模型实验结果有关键影响的超参数, 包括相似性融合系数、训练样本数量以及损失权重。

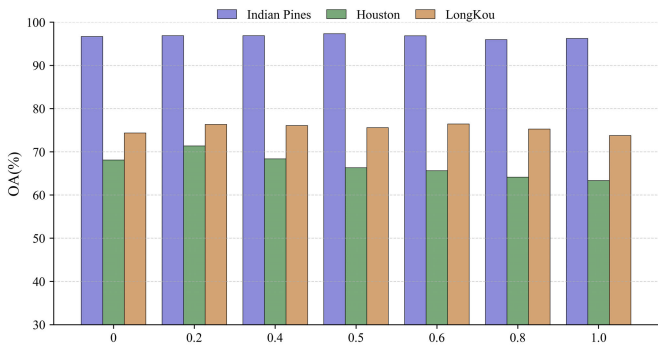


图11. 不同光谱相似性融合系数下三个数据集的实验结果。

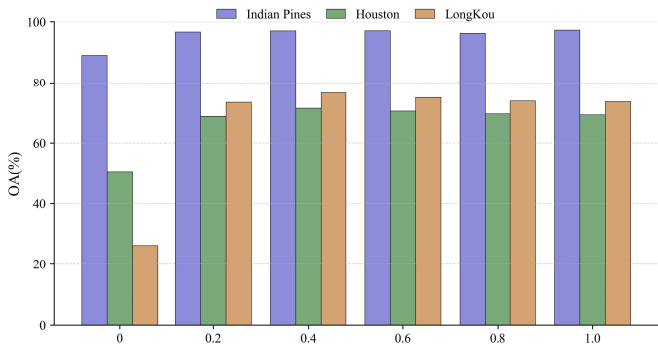


图12. 不同损失融合权重下三个数据集的实验结果。

1) 光谱相似性融合系数 λ_{sim} : 该系数用于调整光谱-文本相似度矩阵 S^{spe} 与空间-文本相似性矩阵 S^{spa} 之间的线性融合比例, 其值为 $\lambda_{\text{sim}} \in [0, 1]$ 。为了评估该超参数对模型性能的影响, 在消融分析中将 λ_{sim} 设置为七个值: [0, 0.2、0.4、0.5、0.6、0.8、1], 同时保持其他参数处于最优配置以消除混杂因素。如图 11 所示, 根据实验结果, 模型在休斯顿、IP 和 WHHL数据集上分别于 λ_{sim} 值为 0.2、0.5 和 0.6 时达到最优分类性能。该结果表明, 跨模态相似性的融合权重对模型性能有显著影响, 且最优值因数据集而异。

TABLE VII
ABLATION STUDY RESULTS (%) ON THE IP DATASETS, HOUSTON DATASETS, AND WHHL DATASETS WITH ALL CLASSES UNDER ZERO-SHOT SETTINGS

Method	Setting	IP			Houston			WHHL		
		OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Baseline	Setting 1	93.35	76.90	75.79	67.63	66.99	49.78	73.60	63.24	54.08
w/o SPE module	Setting 2	94.41	80.76	80.05	68.04	65.97	49.52	74.86	66.94	55.70
w/o SSUE module	Setting 3	96.71	81.02	85.03	68.11	68.32	50.81	74.37	63.38	55.24
Full	Setting 4	97.34	81.45	87.78	71.36	70.35	55.79	76.46	67.74	59.48

- 1) *Setting 1*: Only baseline model, with no spectral–text cross-modal alignment space and no SSEB for feature extraction.
- 2) *Setting 2 (Ablation of SPE Module)*: Replace the SPE text with the original class name. This setting is used to evaluate the contribution of rich textual prompts in representing visual features.
- 3) *Setting 3 (Ablation of SSUE Module)*: Set similarity fusion coefficient λ_{sim} to 0. This setting is used to analyze the contribution of the spectral feature extraction capability in the SSUE module to model performance.
- 4) *Setting 4*: Full SpectralZero model.

The results of the ablation experiments are shown in Table VII. Based on these results, the following conclusions can be drawn.

- 1) Comparing the Setting 4 with the Setting 2, we observe improvements in OA of 2.93%, 3.32%, and 1.60%, and improvements in AA of 0.69%, 4.38%, and 0.80% on the IP, Houston, and WHHL datasets, respectively. These results demonstrate that semantic prompt-enhanced texts play a significant role in improving the model’s ability to express the visual characteristics of land-cover types and effectively facilitate semantic alignment between image and text features.
- 2) Comparing the Setting 4 with the Setting 3, OA is improved by 0.63%, 3.32%, and 2.09%, AA is improved by 0.43%, 2.03%, and 4.36% on the IP, Houston, and WHHL datasets, respectively. Overall, these results validate the effectiveness of the SSUE module in constructing semantic alignment in the spectral dimension, which is particularly evident in datasets with high spectral discriminability.
- 3) Compared with Setting 1, Setting 4 achieves significant performance improvements on all three datasets, with OA improvements of 3.99% (IP), 3.73% (Houston), and 2.86% (WHHL), and AA improvements of 4.55% (IP), 3.36% (Houston), and 4.50% (WHHL). These results fully demonstrate that the proposed dual-branch alignment similarity weighting fusion strategy can effectively enhance the model’s discriminative power and generalization performance in HSI zero-shot classification tasks.
- 2) *Hyperparameter Analysis*: This section presents an analysis of several key hyperparameters that have a critical impact on the experimental results of the model, including the similarity fusion coefficient, the number of training samples, and the loss weights.

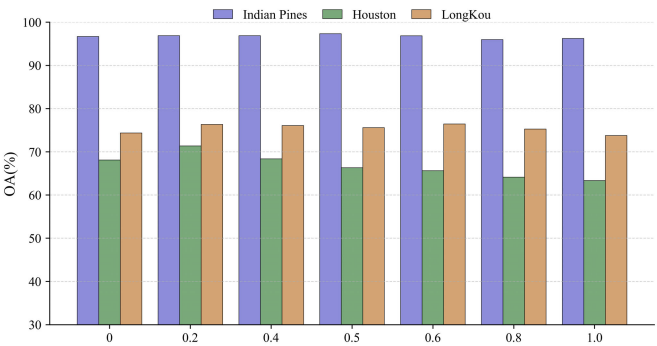


Fig. 11. Experimental results on the three datasets under different spectral similarity fusion coefficients.

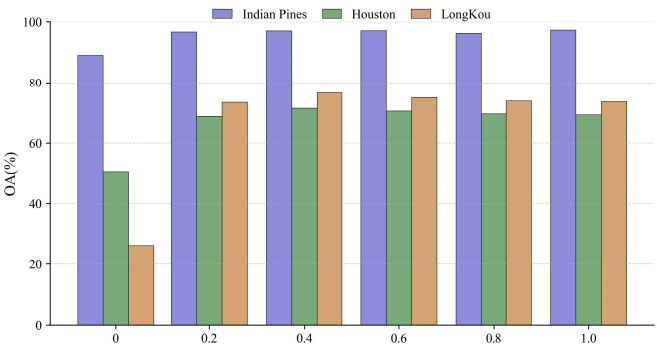


Fig. 12. Experimental results on the three datasets under different loss fusion weights.

- 1) *Spectral Similarity Fusion Coefficient λ_{sim}* : This coefficient is used to adjust the linear fusion ratio between the spectral–text similarity matrix S^{spe} and the spatial–text similarity matrix S^{spa} , with $\lambda_{\text{sim}} \in [0, 1]$. To evaluate the impact of this hyperparameter on model performance, we set λ_{sim} to seven values: [0, 0.2, 0.4, 0.5, 0.6, 0.8, 1] for ablation analysis, while keeping other parameters at their optimal configurations to eliminate confounding factors. As shown in Fig. 11, according to the experimental results, the model achieves optimal classification performance at λ_{sim} values of 0.2, 0.5, and 0.6 on the Houston, IP, and WHHL datasets, respectively. This result indicates that the fusion weighting of cross-modal similarity has a significant impact on the model’s performance, and that the optimal value differs across datasets.

表VIII SSUE超参数设置

Dataset	Image Encoder	MaxPool	SOEB Conv3d	SOEB Conv1d	MLP	Projection Dim
IP	Kernel (3, 3, 3) Padding (0, 1, 0) Bias True	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (4, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 4 Padding (1,) Kernel (3,)	Output Dim 128	768
Houston	Kernel (3, 1, 1) Padding (1, 0, 0) Bias True	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (1, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 16 Padding (1,) Kernel (3,)	Output Dim 128	768
WHHL	Kernel (3, 3, 3) Padding (1, 0, 0) Bias False	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (4, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 8 Padding (1,) Kernel (3,)	Output Dim 128	768

2) 损失融合权重 λ_{loss} : 该参数用于调整总损失函数中 InfoNCE 损失与辅助损失（交叉熵损失）之间的线性加权比例。通过调节 λ_{loss} 的值，可以平衡不同损失项的贡献，防止某一损失项主导优化过程，从而抑制其他监督信号的优化效果。为分析该参数对模型性能的影响，我们将 λ_{loss} 设置为六个值 [0, 0.2, 0.4, 0.6,0.8, 1] 用于分析。实验结果如图 12所示。根据结果，模型在休斯顿、IP 和 WHHL数据集上分别于 λ_{loss} 值为 0.4、0.4 和 1 时达到最优性能。这些结果表明，不同损失项之间的协调对模型的训练效率和最终性能均有显著影响。

3) 其他参数：除了主要的超参数设置外，模型实现中还配置了若干结构参数，如表VIII所示，包括三维卷积的卷积核大小、偏置项的使用以及最大池化操作的步长设置。在休斯顿数据集上，三维卷积的卷积核大小设置为(3, 1, 1)，对应的填充为(1, 0, 0)，并且在训练过程中启用偏置项。对于IP和WHHL数据集，统一采用(3, 3, 3)的卷积核大小和(1, 1, 1)的填充。关于最大池化操作，休斯顿数据集使用(1, 2, 2)的池化步长以保留更多光谱信息，而IP和WHHL数据集则使用(4, 2, 2)的步长以进一步压缩空间维度并提高处理效率。最后，投影维度表示将光谱分支和空间分支特征映射到共享嵌入空间的投影头的输出维度，所有数据集均设为768。

IV. 结论

本文提出了一种用于零样本分类的高光谱图像分类方法，名为SpectralZero，旨在充分挖掘高光谱图像中的语义信息，并增强

模型对不可见类的识别能力。为了实现语义知识的有效迁移，我们引入了SPE模块，该模块将原始类别标签扩展为带有视觉属性的自然语言描述。这些描述作为迁移的语义锚，有效弥合了文本特征与图像特征之间的语义鸿沟，并提供了一种更具应用导向的表达方式。为了联合建模图像的光谱特征和空间特征，我们设计了SSUE模块，该模块显式地将光谱特征从混合特征中解耦出来进行独立提取与投影，同时利用空间特征进行辅助分类和对齐。这两类视觉特征被映射到两个独立的嵌入空间中，并与语义特征对齐。最后，相似性矩阵的线性加权融合引导了跨模态对比学习过程。实验结果表明，与现有方法相比，所提出的框架在三个公开的高光谱图像数据集上取得了更优的分类性能，验证了其有效性和泛化能力。

在未来的工作中，我们将进一步探索更实际的场景，例如将模型扩展到现实世界中土地覆盖边界模糊且不可见类不可预测的广义零样本分类任务。我们还将专注于提升跨数据集泛化能力，以更好地满足高光谱图像应用对可迁移性的需求。

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TABLE VIII
HYPERPARAMETER SETTINGS FOR SSUE

Dataset	Image Encoder	MaxPool	SOEB Conv3d	SOEB Conv1d	MLP	Projection Dim
IP	Kernel (3, 3, 3) Padding (1, 0, 0) Bias True	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (4, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 4 Padding (1,) Kernel (3,)	Output Dim 128	768
Houston	Kernel (3, 1, 1) Padding (1, 0, 0) Bias True	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (1, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 16 Padding (1,) Kernel (3,)	Output Dim 128	768
WHHL	Kernel (3, 3, 3) Padding (1, 0, 0) Bias False	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (4, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 8 Padding (1,) Kernel (3,)	Output Dim 128	768

- 2) *Loss Fusion Weight λ_{loss}* : This parameter is used to adjust the linear weighting ratio between the InfoNCE loss and the auxiliary loss (cross-entropy loss) in the total loss function. By tuning the value of λ_{loss} , it is possible to balance the contributions of different loss terms and prevent one loss term from dominating the optimization process, thereby suppressing the optimization effect of other supervisory signals. To analyze the effect of this parameter on model performance, we set λ_{loss} to six values [0, 0.2, 0.4, 0.6, 0.8, 1] for analysis. The experimental results are shown in Fig. 12. According to the results, the model achieves optimal performance at λ_{loss} values of 0.4, 0.4, and 1 on the Houston, IP, and WHHL datasets, respectively. These results indicate that the coordination between different loss terms has a significant impact on both the training efficiency and final performance of the model.
- 3) *Other Parameters*: In addition to the main hyperparameter settings, several structural parameters are also configured in the model implementation, which are provided in Table VIII, including the kernel size for 3D convolution, the use of bias terms, and the stride settings for max-pooling operations. On the Houston dataset, the 3D convolution kernel size is set to (3, 1, 1) with corresponding padding of (1, 0, 0), and bias terms are enabled during training. For both the IP and WHHL datasets, a unified kernel size of (3, 3, 3) and padding of (1, 1, 1) are adopted. Regarding max-pooling operations, a pooling stride of (1, 2, 2) is used for the Houston dataset to preserve more spectral information, while a stride of (4, 2, 2) is used for the IP and WHHL datasets to further compress the spatial dimension and improve processing efficiency. Finally, projection dim denotes the output dimensionality of the projection heads that map the spectral and spatial branch features into the shared embedding space, which is set to 768 for all datasets.

IV. CONCLUSION

This article proposes an HSI classification method for zero-shot classification which called SpectralZero, aiming to fully exploit the semantic information in HSI and enhance the

model’s recognition ability for unseen classes. To achieve effective transfer of semantic knowledge, we introduce the SPE module, which extends the original class labels into natural language descriptions with visual attributes. These serve as semantic anchors for transfer, effectively bridging the semantic gap between textual and image features, and providing a more application-oriented mode of expression. To jointly model the spectral and spatial features of images, we design the SSUE module, which explicitly decouples spectral features from mixed features for independent extraction and projection, while utilizing spatial features for auxiliary classification and alignment. The two types of visual features are mapped into two independent embedding spaces and aligned with semantic features. Finally, a linear weighted fusion of similarity matrices guides the cross-modal contrastive learning process. Experimental results demonstrate that the proposed framework achieves superior classification performance compared to existing methods on three publicly available HSI datasets, verifying its effectiveness and generalizability.

In future work, we will further explore more practical settings, such as extending the model to the generalized zero-shot classification task in real-world scenarios where land-cover boundaries are ambiguous and unseen classes are unpredictable. We will also focus on improving cross-dataset generalization to better meet the transferability requirements of HSI applications.

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他的研究兴趣涵盖计算机视觉、机器学习、高光谱图像分类、小样本学习和零样本学习。



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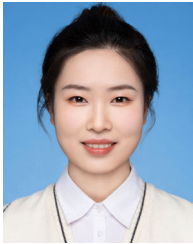
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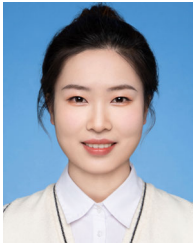
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